Model Compression

Danna Gurari
University of Colorado Boulder
Spring 2022



https://home.cs.colorado.edu/~DrG/Courses/NeuralNetworksAndDeepLearning/AboutCourse.html

Review

- Last lecture topic:
  - Multi-task learning
  - Few-shot learning
  - Zero-shot learning
- Assignments
  - Final project proposal due Friday
- Questions?

Today's Topics

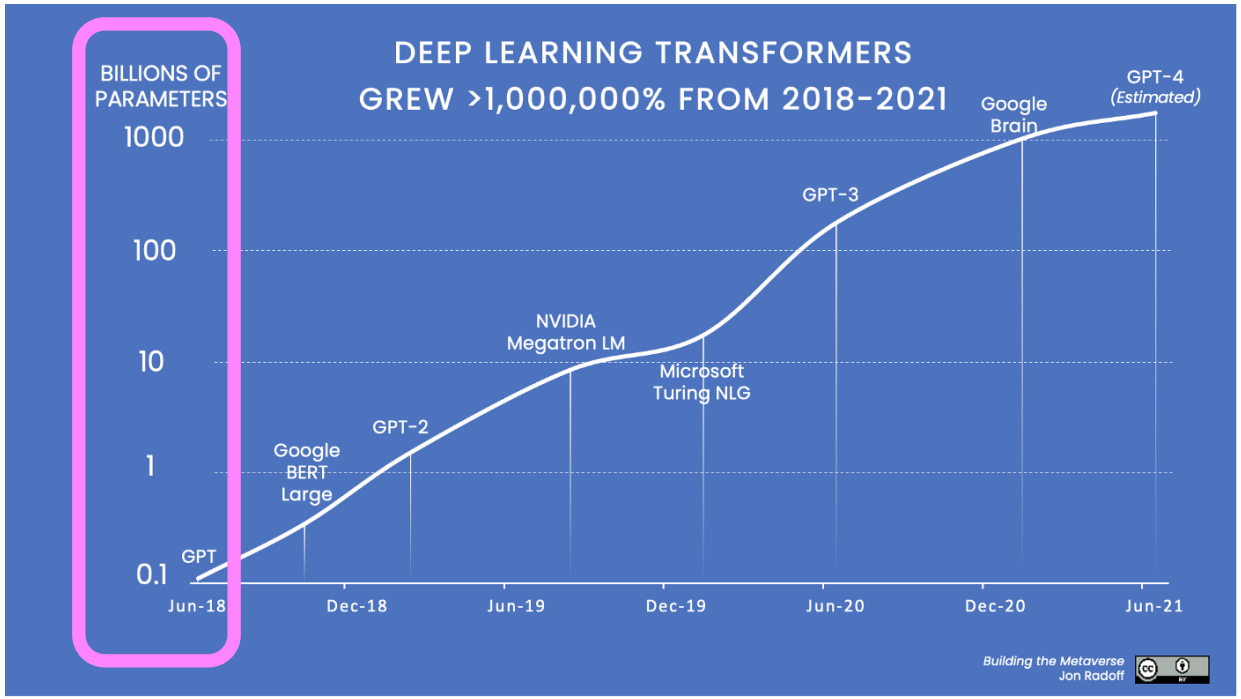
- Motivation
- Key idea: knowledge distillation
- Knowledge distillation for CNNs (vision problems)
- Knowledge distillation for Transformers (language problems)
- Latex tutorial

Today's Topics

- Motivation
- Key idea: knowledge distillation
- Knowledge distillation for CNNs (vision problems)
- Knowledge distillation for Transformers (language problems)
- Latex tutorial

Trend: Parameter-Heavy Models

Language – pretrained transformers



https://medium.com/building-the-metaverse/the-metaverse-and-artificial-intelligence-ai-577343895411

Vision – ImageNet classification

Architecture	Year	Top-1 Accuracy	# Parameters
DenseNet-169	2017	76.2%	14M
Inception-v3	2016	78.8%	24M
Inception-resnet-v2	2017	80.1%	56M
PolyNet	2017	81.3%	92M
SENet	2018	82.7%	146M
GPipe	2018	84.3%	557M
ResNeXt-101 32x48d	2019	85.4%	829M

Modern Neural Networks Are a Mismatch for Many Real-World Applications



https://www.ephotozine.com/article/19-things-to-look-out-for-in-a-smartphone-camera--31055



https://en.wikipedia.org/wiki/Wearable_technology



https://www.buzzfeednews.com/article/katienotopoulos/facebook-is-making-camera-glasses-ha-ha-oh-no

Modern Neural Networks Are a Mismatch for Many Real-World Applications

- Large inference time (i.e., incompatible for real-time applications)
- Large memory footprint (e.g., incompatible with limited memory on edge devices)
- Large computational cost (e.g., incompatible with limited battery on edge devices)
- Potential for large environmental costs

Idea: develop compact models so deep learning models can be used more efficiently and for more applications

Today's Topics

- Motivation
- Key idea: knowledge distillation
- Knowledge distillation for CNNs (vision problems)
- Knowledge distillation for Transformers (language problems)
- Latex tutorial

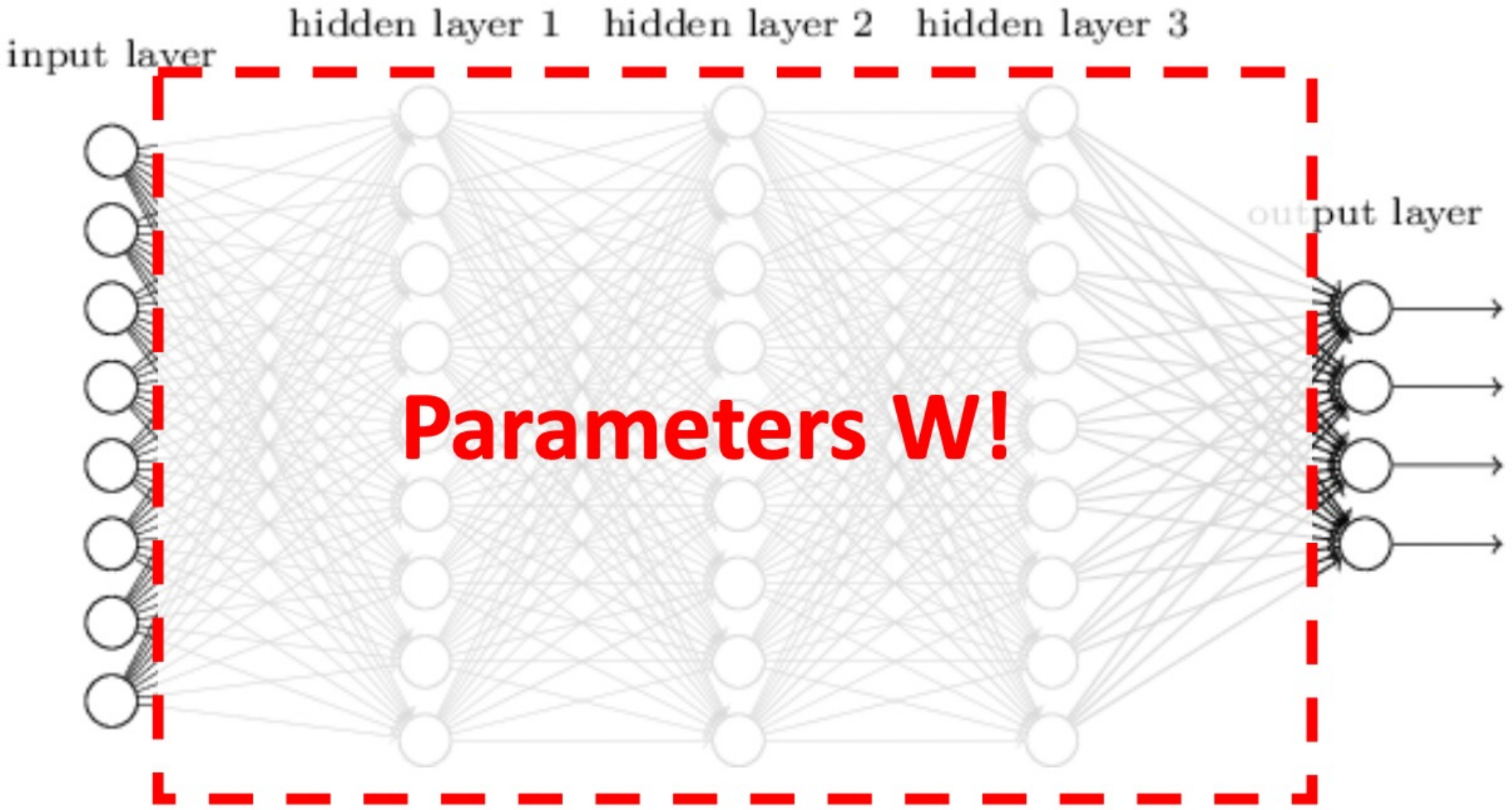
Popular Approach: Knowledge Distillation



A student learns from a knowledgeable teacher

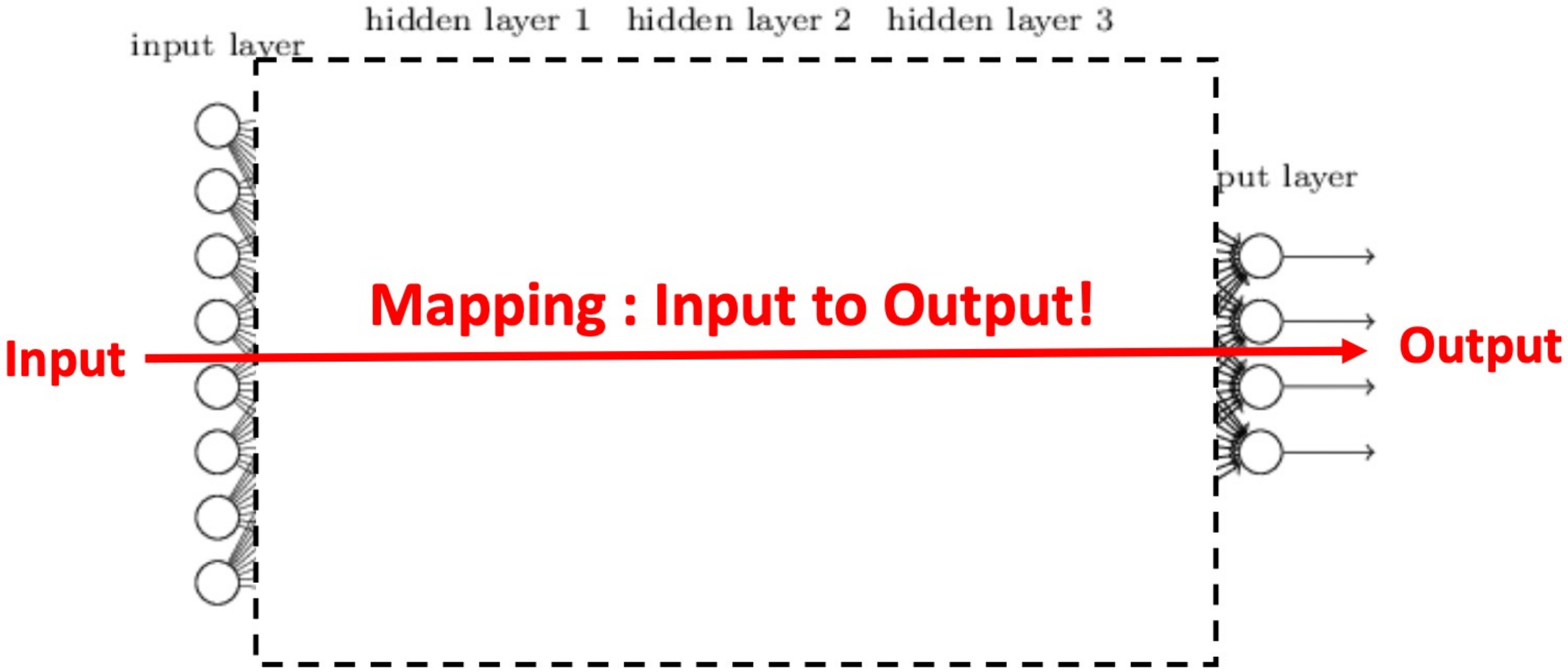
Image source: https://www.waterford.org/education/teacher-student-relationships/

Key Question: What is Knowledge?



http://ir.hit.edu.cn/~xiachongfeng/slides/Knowledge%20Distillation.pdf

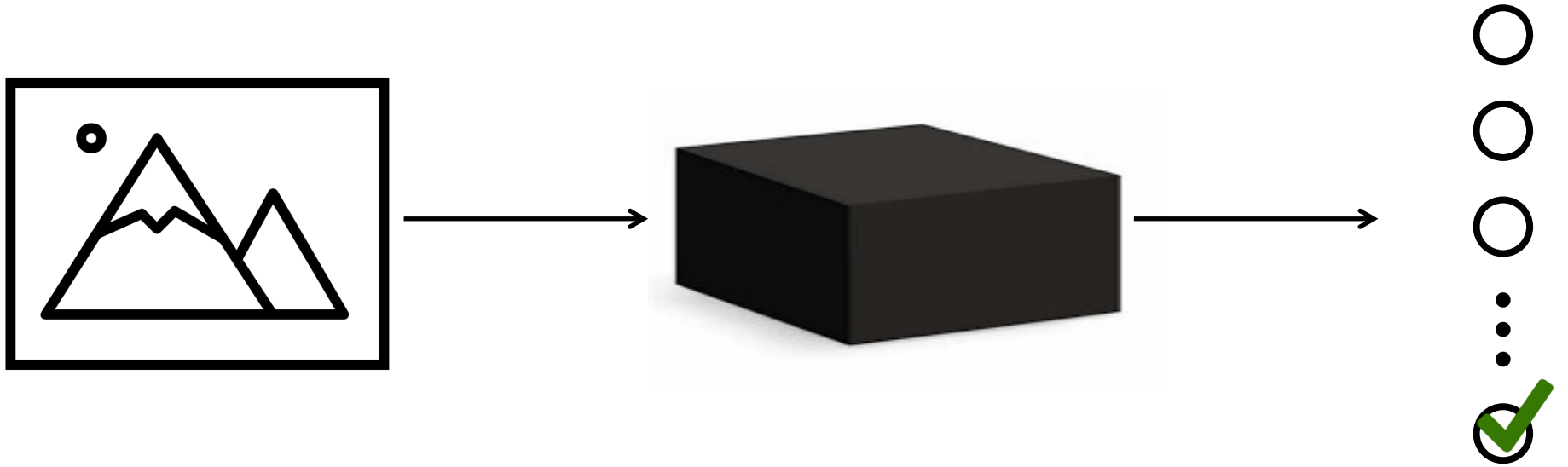
Knowledge Is: Input to Output Mapping



http://ir.hit.edu.cn/~xiachongfeng/slides/Knowledge%20Distillation.pdf

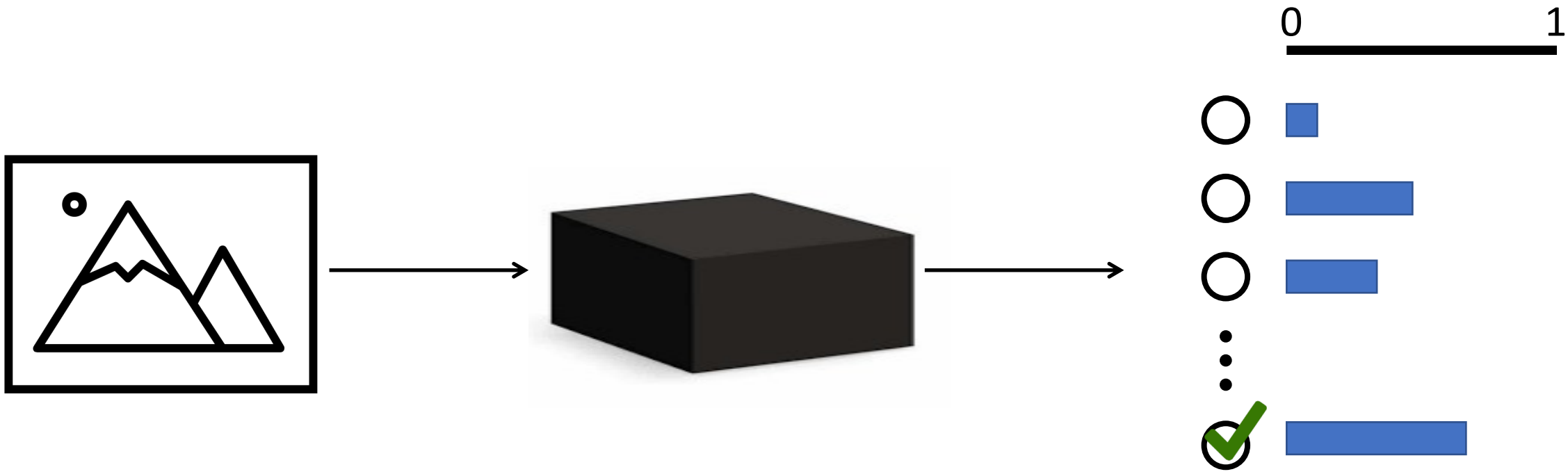
Knowledge Is: Input to Output Mapping

Target mapping: ground truth (1-hot vector)



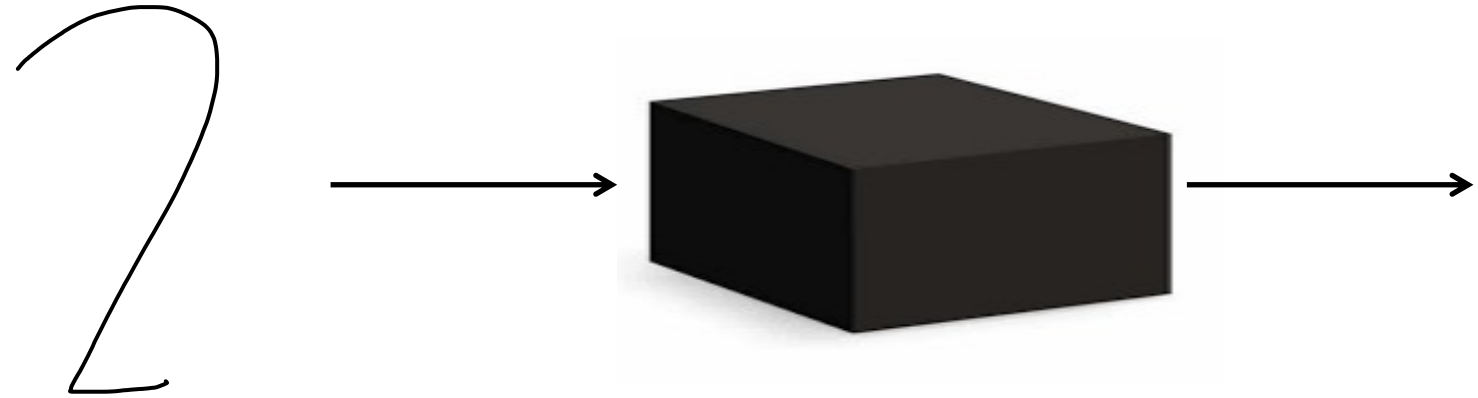
Knowledge Is: Input to Output Mapping

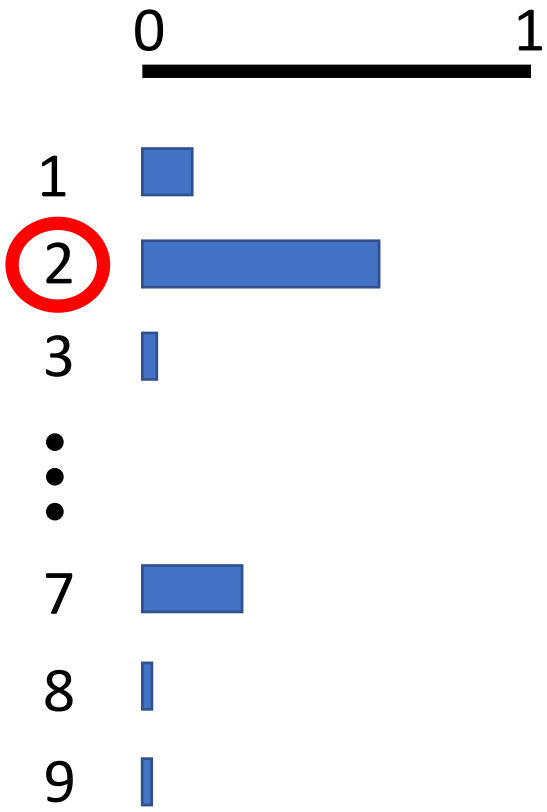
Target mapping: probability distribution from a model offers
further insights into similarities and differences of categories



Knowledge Is: Input to Output Mapping

Target mapping: probability distribution from a model offers
further insights into similarities and differences of categories
- Attempts to identify ground truth category
- Also, shares that 2 has similar characteristics to 7 and 1



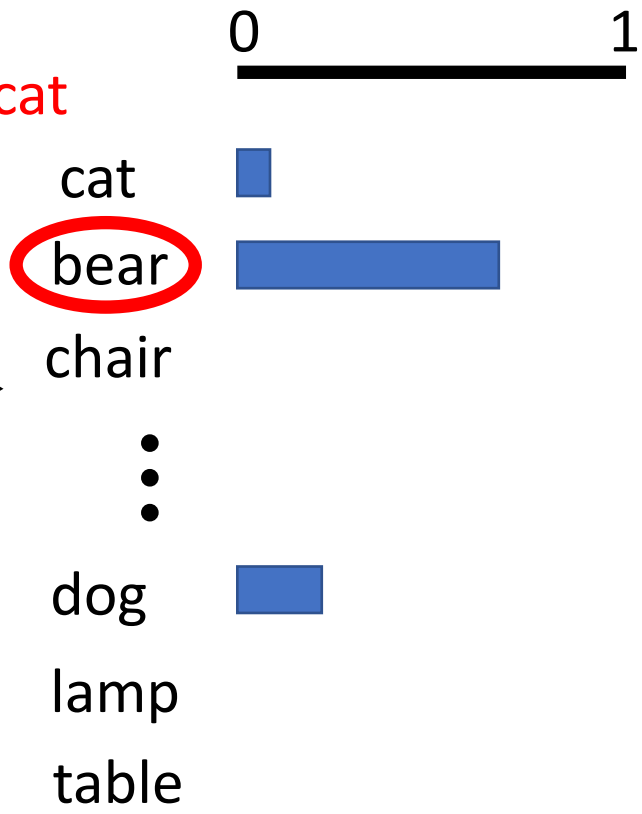


Knowledge Is: Input to Output Mapping

Target mapping: probability distribution from a model offers
further insights into similarities and differences of categories
- Attempts to identify ground truth category
- Also, shares that bear has similar characteristics to dog and cat







Knowledge Is: Input to Output Mapping

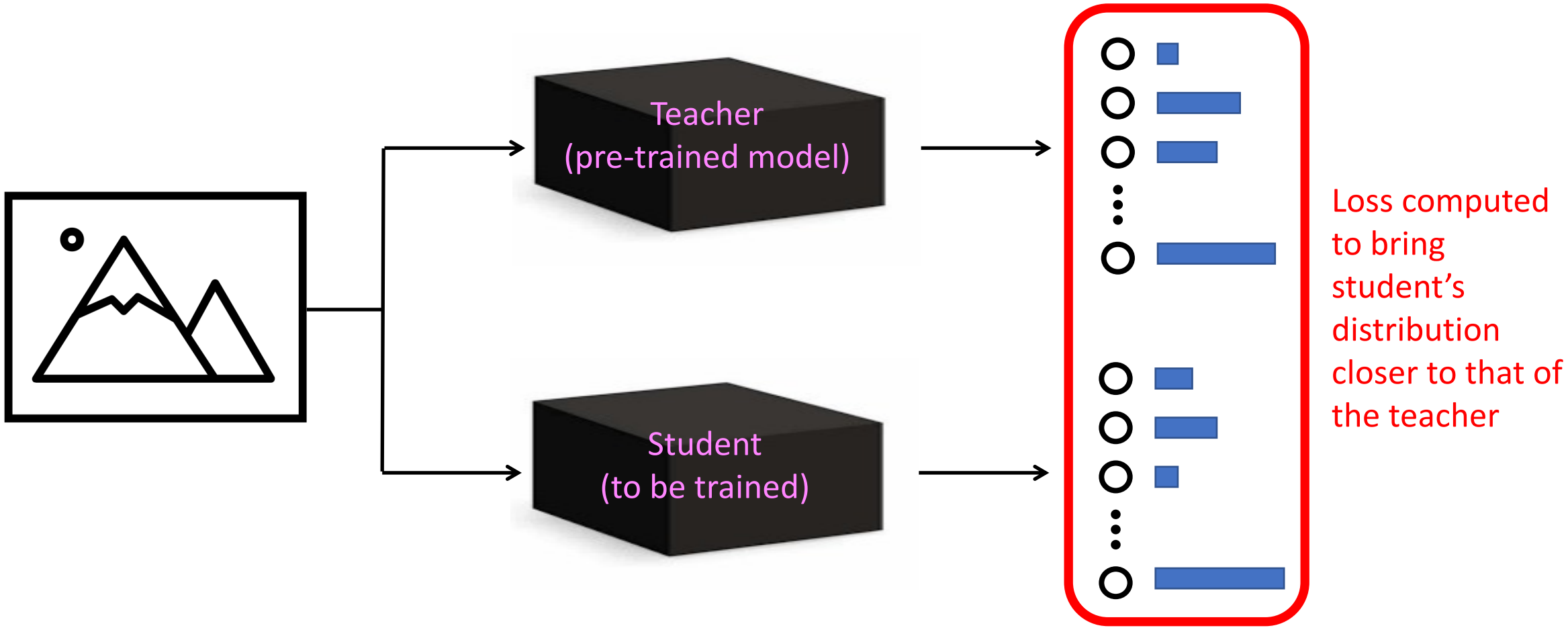
Target mapping: probability distribution from a model offers further insights into similarities and differences of categories

- Attempts to identify ground truth category
- Also, shares that bear has similar characteristics to dog and cat

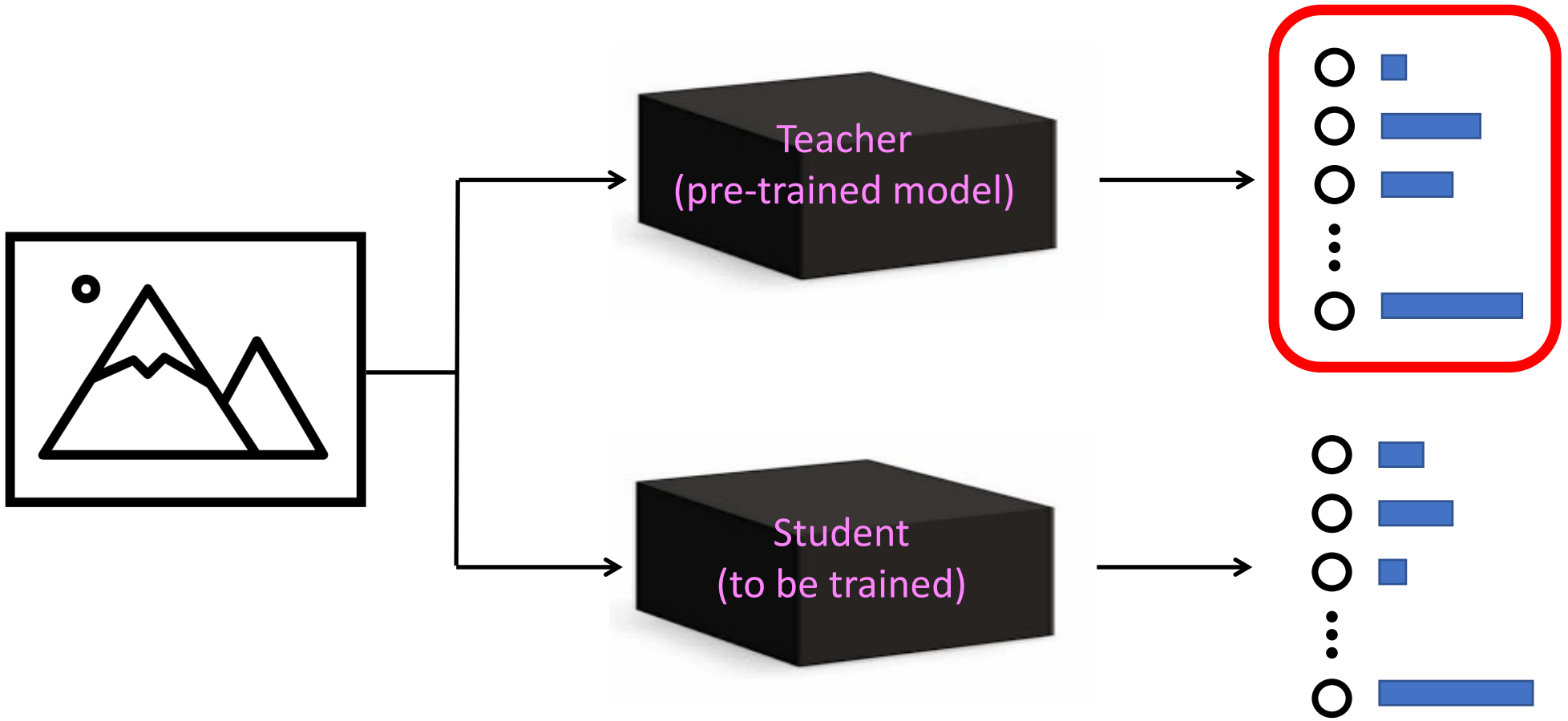
Idea: not only teach a compact student about the ground truth, but also teach it about the relationships between categories

Hinton, Vinyals, and Dean. Distilling the knowledge in a neural network. *arXiv* 2015.

Knowledge Distillation: Teach Student the "Dark Knowledge" of Teacher

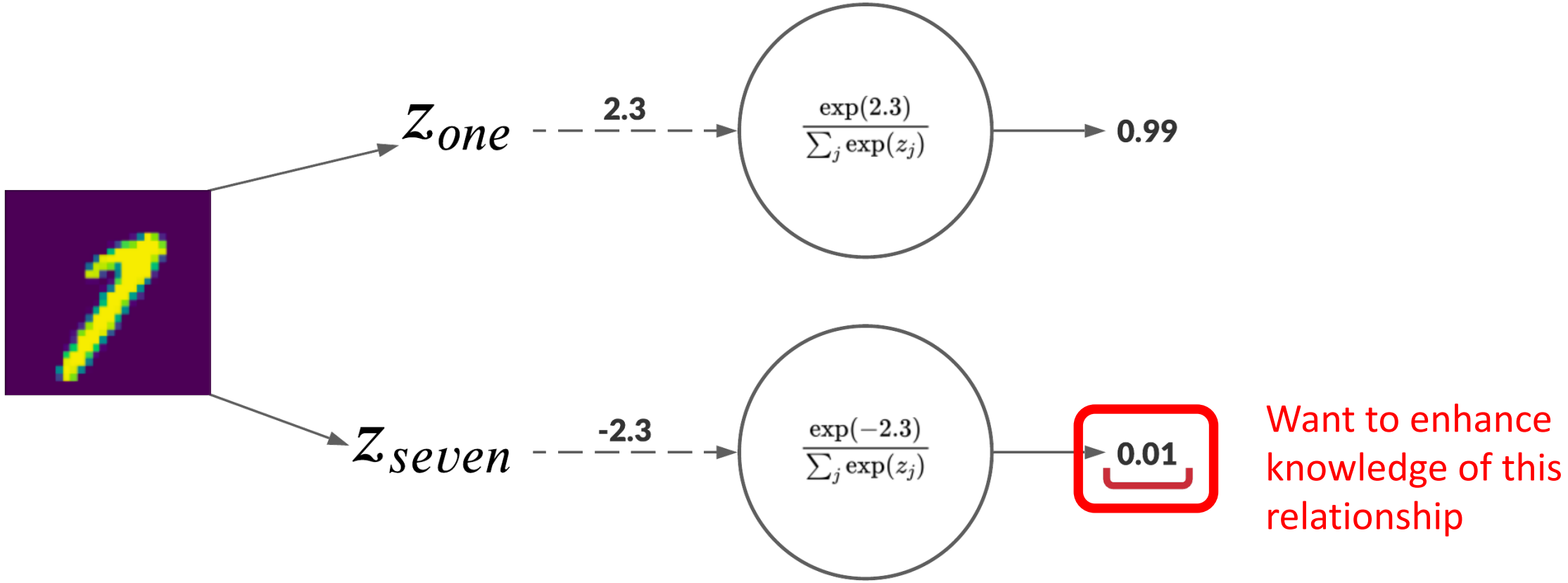


Knowledge Distillation: Teach Student the "Dark Knowledge" of Teacher



Knowledge Distillation: Rebalance ("Soften") Probability Distribution Across Categories

Recall Softmax: converts vector of scores into a probability distribution that sums to 1



https://wandb.ai/authors/knowledge-distillation/reports/Distilling-Knowledge-in-Neural-Networks--VmlldzoyMjkxODk

Knowledge Distillation: Rebalance ("Soften") Probability Distribution Across Categories

Recall Softmax: converts vector of scores into a probability distribution that sums to 1

Get rid of negative values while preserving original order of scores

$$\sigma(\mathbf{z})_i = \frac{e^{z_i}}{\sum_{j=1}^{K} e^{z_j}}$$

i = 1, …, K

Number of classes

Divide each node's score by sum of all entries to make them sum to 1 (normalization)

Useful tutorial: https://towardsdatascience.com/exploring-the-softmax-function-578c8b0fb15

Knowledge Distillation: Rebalance ("Soften") Probability Distribution Across Categories

Generalized Softmax: converts vector of scores into a probability distribution that sums to 1 with temperature

$$\sigma(\mathbf{z})_i = \frac{\exp(z_i / T)}{\sum_j \exp(z_j / T)}$$

What is the typical value of T used for softmax?

Idea: set the temperature to a value greater than 1

Knowledge Distillation: Rebalance ("Soften") Probability Distribution Across Categories

Generalized Softmax: converts vector of scores into a probability distribution that sums to 1 with temperature

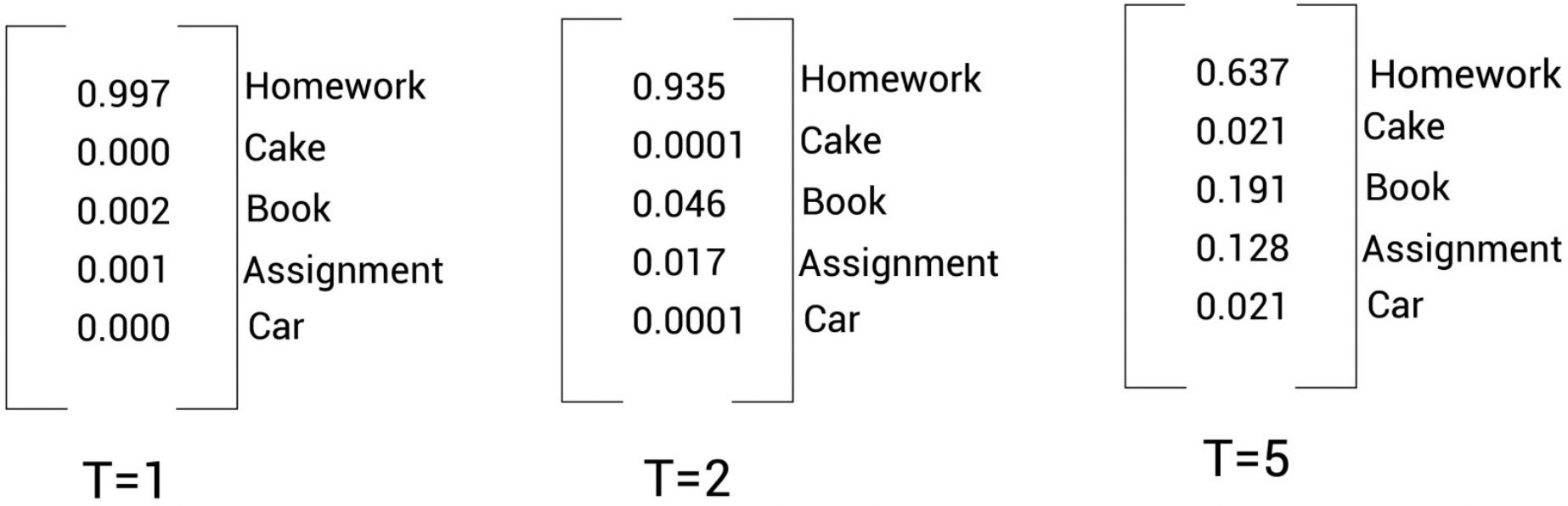
$$\sigma(\mathbf{z})_i = \frac{\exp(z_i / T)}{\sum_j \exp(z_j / T)}$$

Larger T values means more information is available about which categories the teacher found similar to the predicted category

https://wandb.ai/authors/knowledge-distillation/reports/Distilling-Knowledge-in-Neural-Networks--VmlldzoyMjkxODk

Knowledge Distillation: Rebalance ("Soften") Probability Distribution Across Categories

Generalized Softmax: converts vector of scores into a probability distribution that sums to 1 with temperature; e.g.,

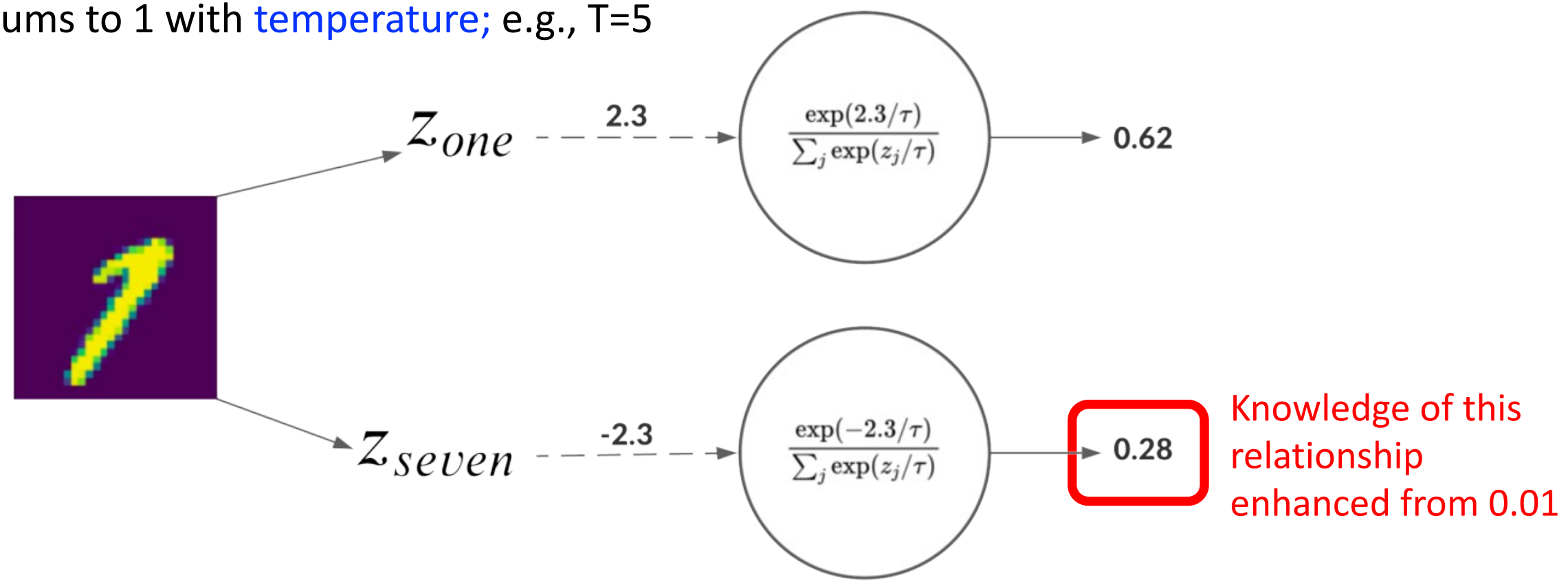


Larger T values means more information is available about which categories the teacher found similar to the predicted category

https://static.packt-cdn.com/downloads/9781838821593_ColorImages.pdf

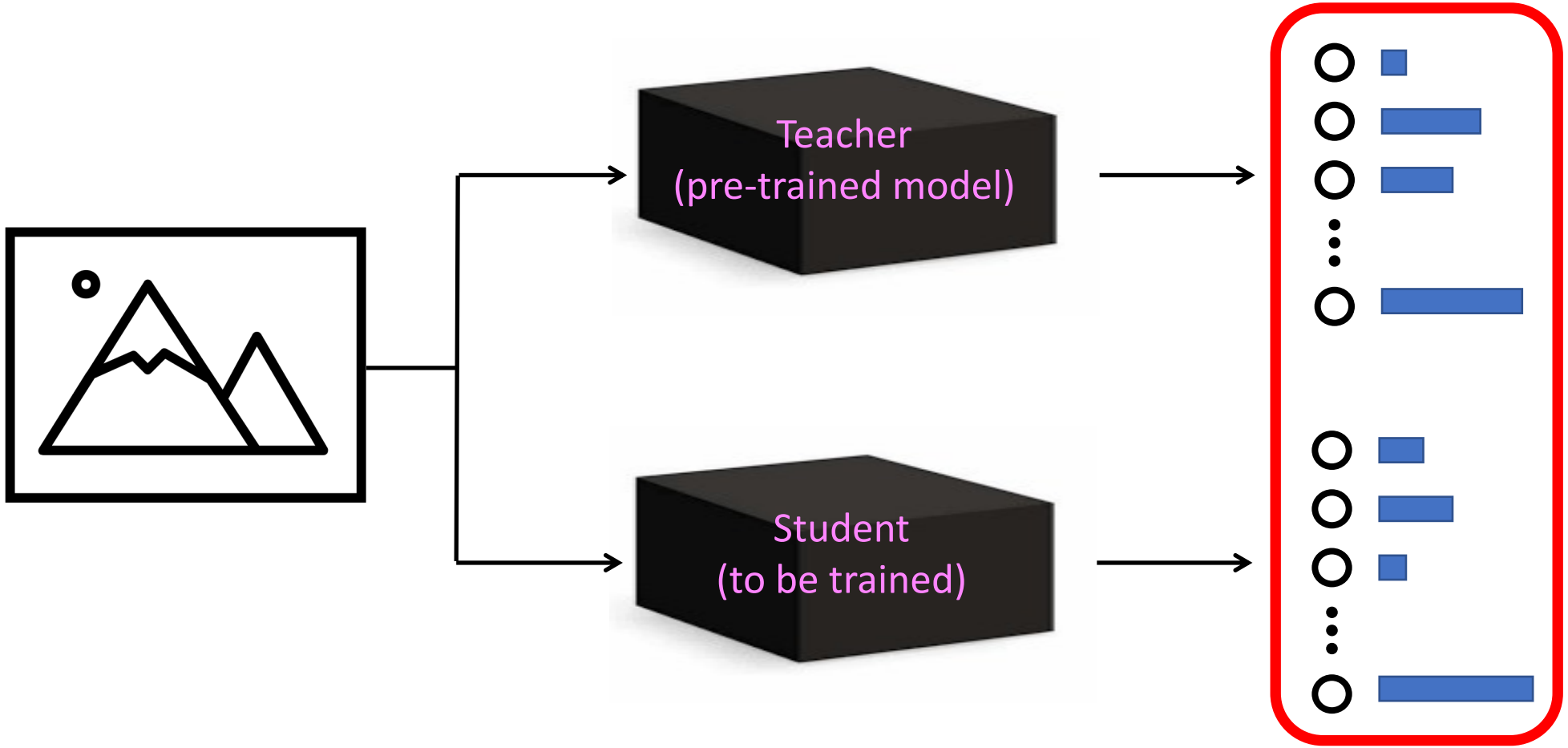
Knowledge Distillation: Rebalance ("Soften") Probability Distribution Across Categories

Generalized Softmax: converts vector of scores into a probability distribution that sums to 1 with temperature; e.g., T=5



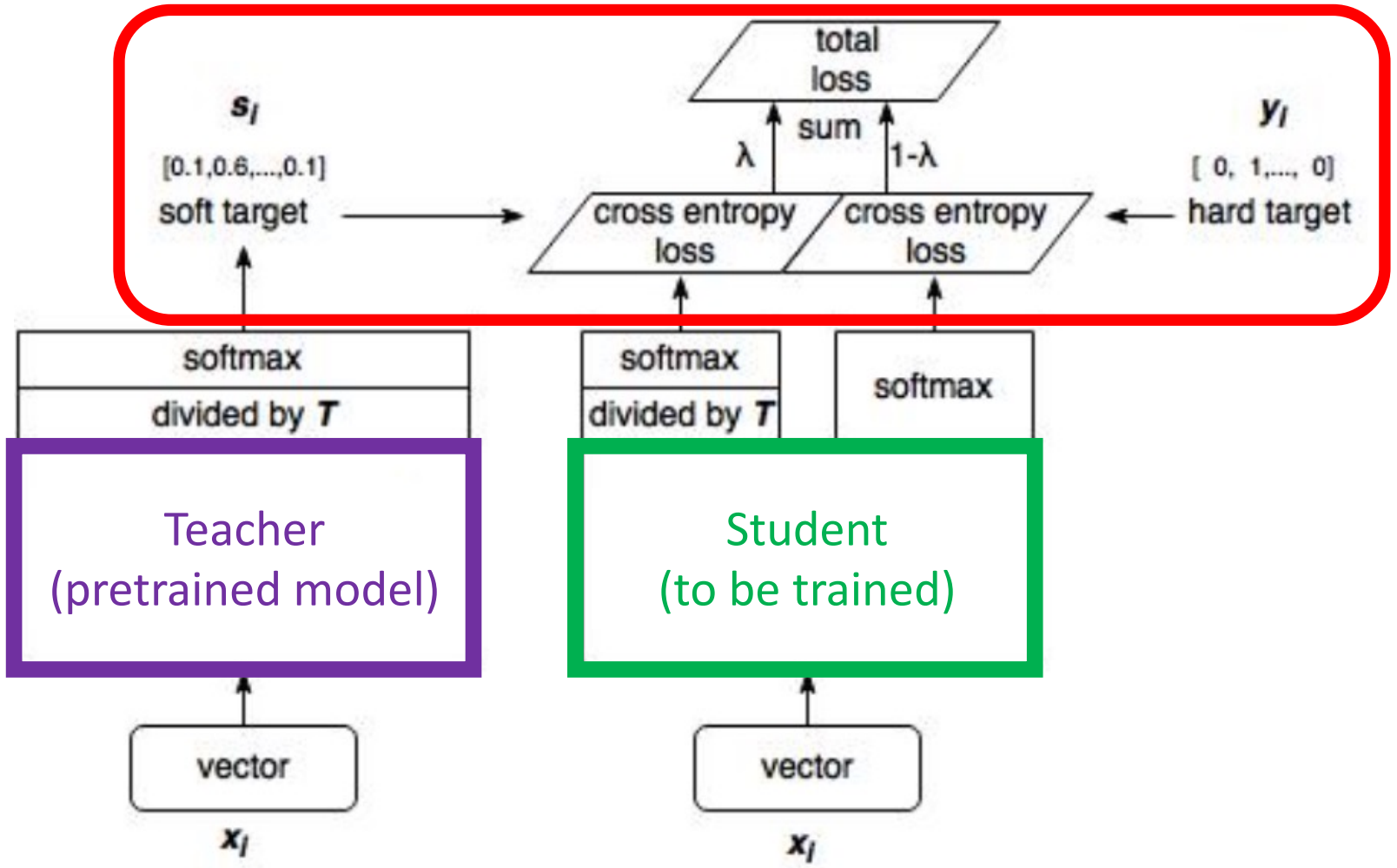
https://wandb.ai/authors/knowledge-distillation/reports/Distilling-Knowledge-in-Neural-Networks--VmlldzoyMjkxODk

Knowledge Distillation: Teach Student the "Dark Knowledge" of Teacher



"Distillation" loss computed to bring student's distribution closer to that of the teacher, using the generalized softmax equation

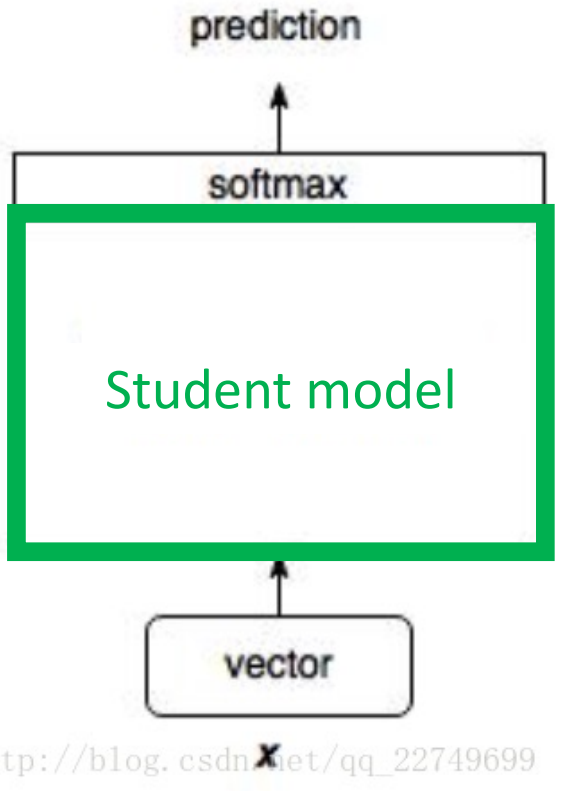
Knowledge Distillation: Teach Student the "Dark Knowledge" of Teacher



Total loss computed during training is a weighted sum of the conventional cross entropy loss and the "distillation loss"

https://blog.csdn.net/qq_22749699/article/details/79460817

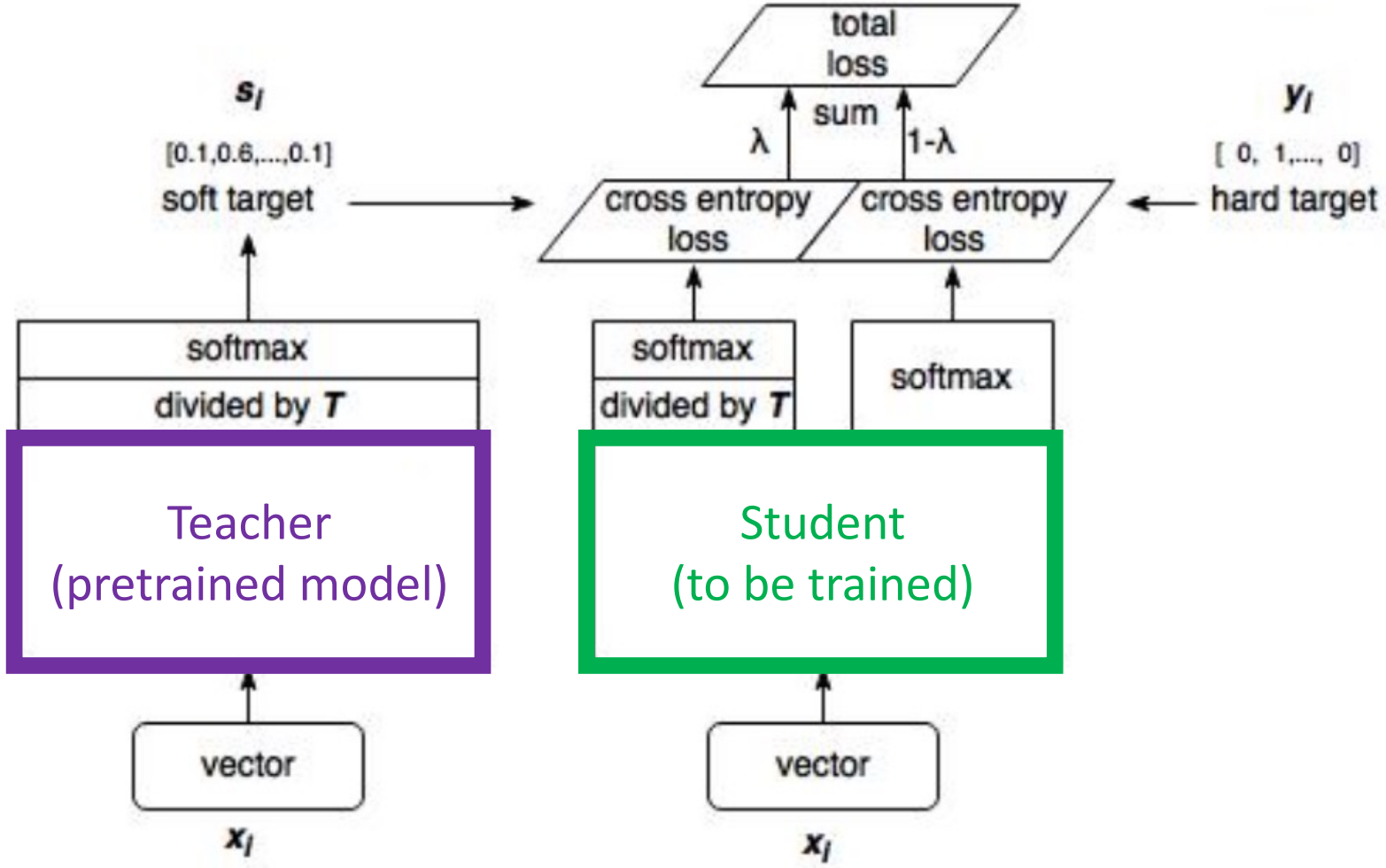
Knowledge Distillation: At Test Time



ttp://blog.csdn.net/qq_22749699

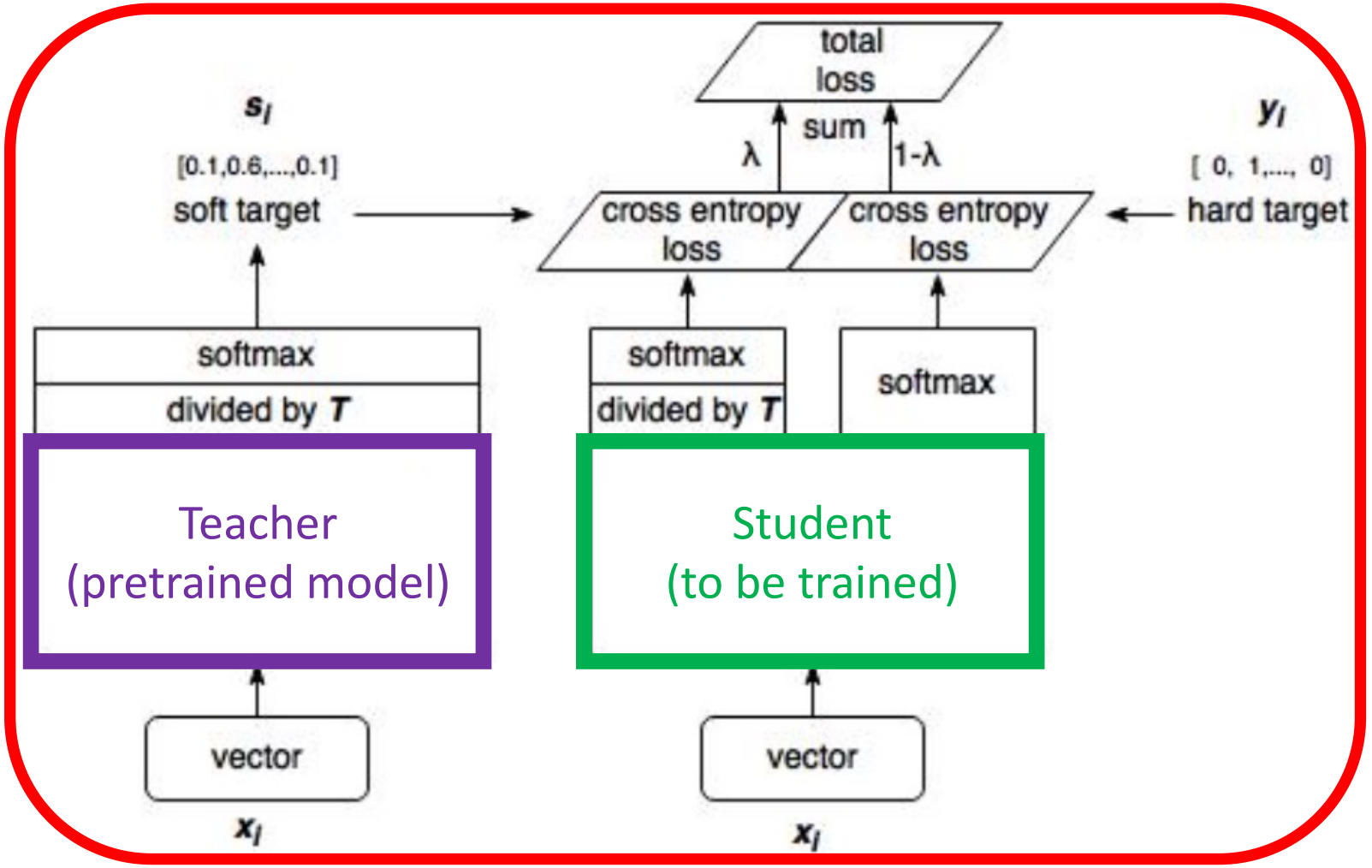
https://blog.csdn.net/qq_22749699/article/details/79460817

Arguably, Any Neural Network Student Could Learn from Any Neural Network Teacher



https://blog.csdn.net/qq_22749699/article/details/79460817

Arguably, Any Neural Network Student Could Learn from Any Neural Network Teacher

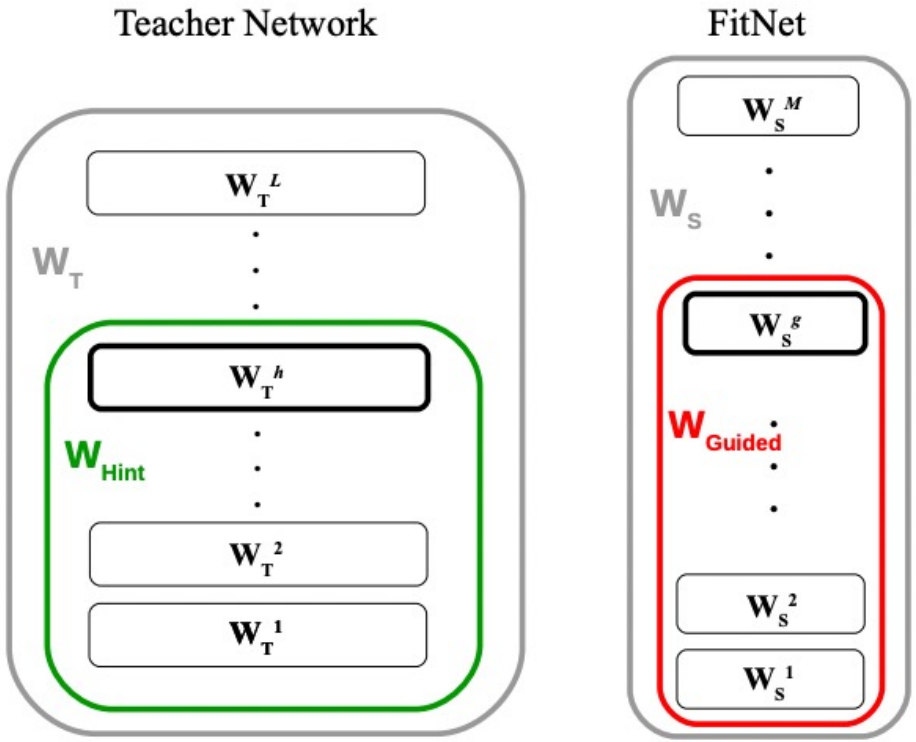


Knowledge distillation is a type of transfer learning

https://blog.csdn.net/qq_22749699/article/details/79460817

Knowledge Distillation Enhancement: Hints

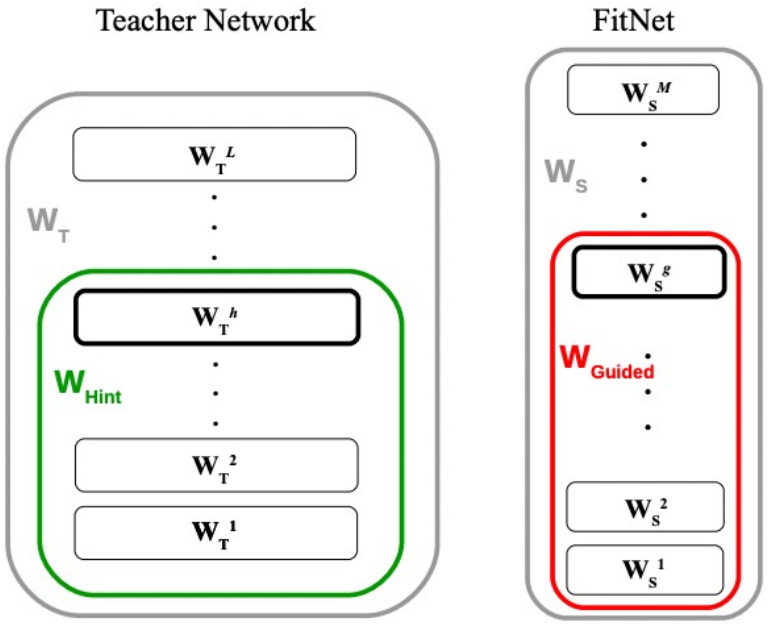
Encourage student (FitNet) to mimic the teacher's feature responses; e.g., output of guided layer should match the output of hint layer



Romero et al. Fitnets: Hints for thin deep nets. ICLR 2015.

Knowledge Distillation Enhancement: Hints

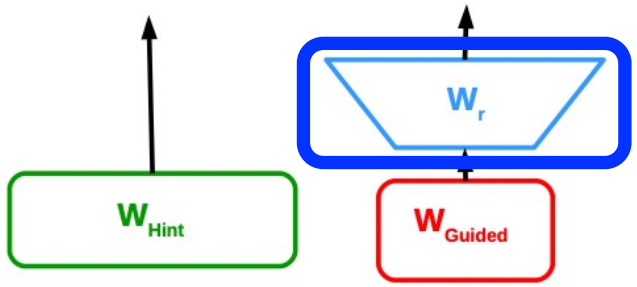
Encourage student (FitNet) to mimic the teacher's feature responses; e.g., output of guided layer should match the output of hint layer



(a) Teacher and Student Networks

Training conducted to learn the intermediate feature

$$W^*_{Guided} = \underset{W_{Guided}}{\operatorname{argmin}} \mathcal{L}_{HT}(W_{Guided}, W_r)$$



Layer added to match size of the hint's output layer

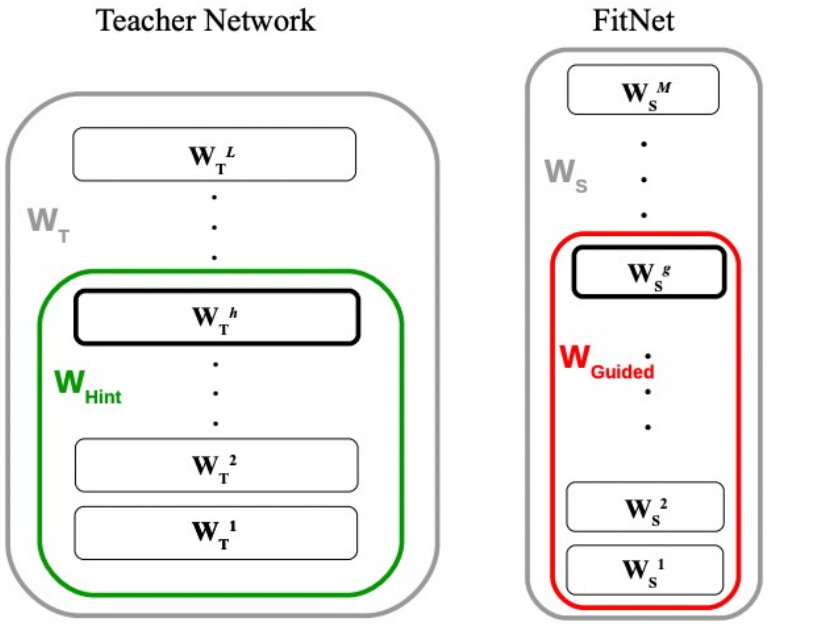
(b) Hints Training

Romero et al. Fitnets: Hints for thin deep nets. ICLR 2015.

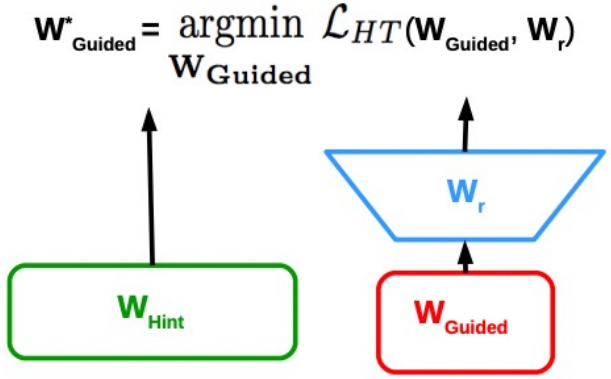
Knowledge Distillation Enhancement: Hints

Encourage student (FitNet) to mimic the teacher's feature responses;
e.g., output of guided layer should match the output of hint layer

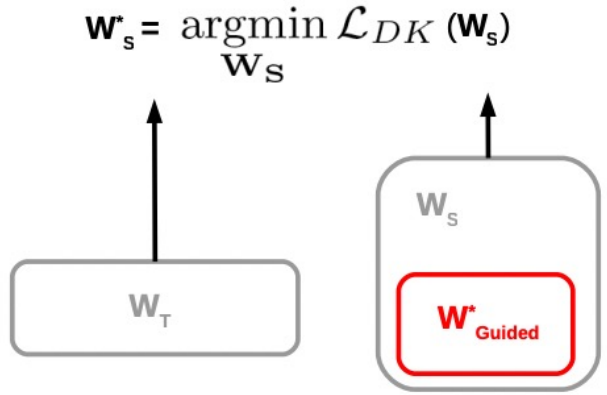
After learning the intermediate features, the whole student network is trained



(a) Teacher and Student Networks



(b) Hints Training



(c) Knowledge Distillation

Romero et al. Fitnets: Hints for thin deep nets. ICLR 2015.

Today's Topics

- Motivation
- Key idea: knowledge distillation
- Knowledge distillation for CNNs (vision problems)
- Knowledge distillation for Transformers (language problems)
- Latex tutorial

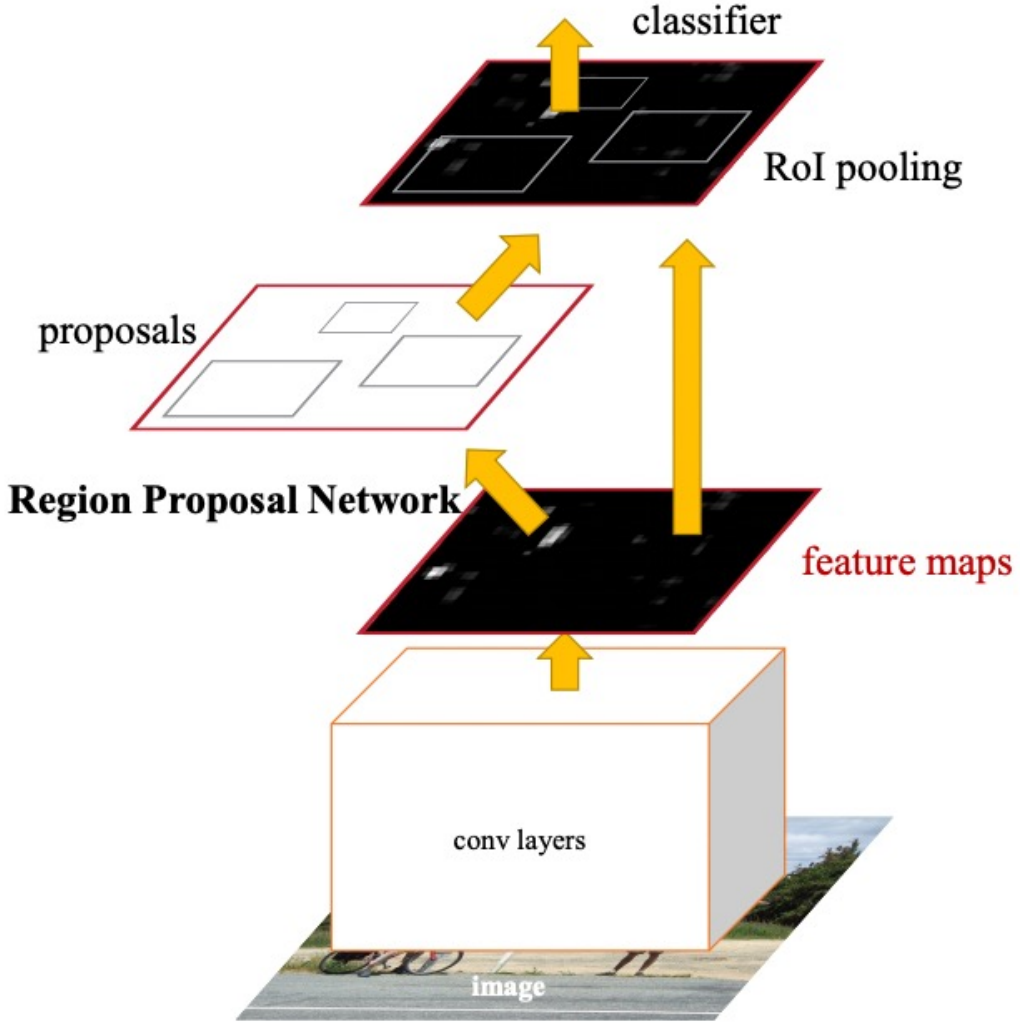
Knowledge Distillation for Vision

- Object detection
- Image classification

Knowledge Distillation for Vision

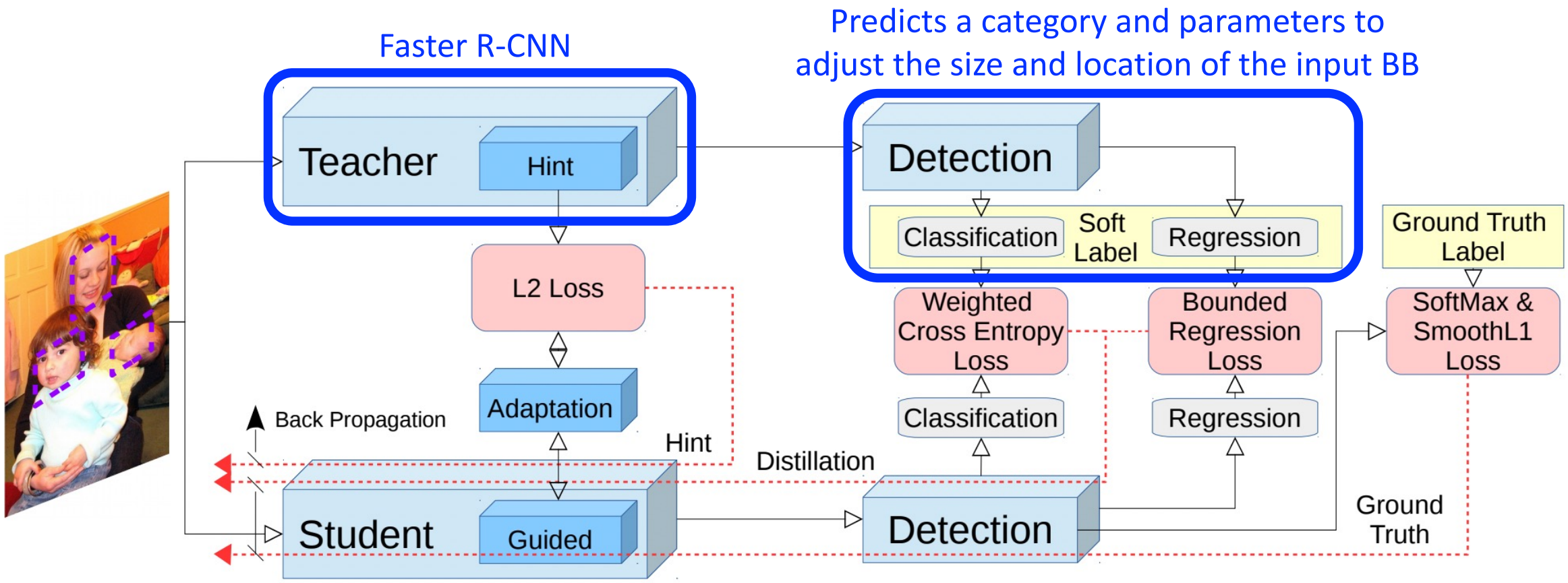
- Object detection
- Image classification

Recall Popular Detection Model: Faster R-CNN



Ren Shaoqing Ren et al. “Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks.” Neurips 2015.

Approach for Creating Compact Student Model



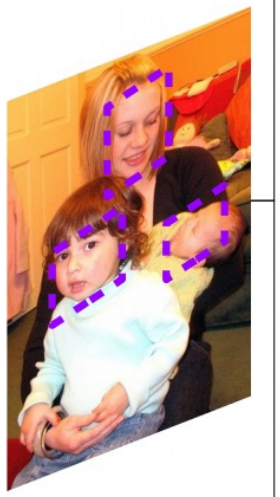
Chen et al. Learning efficient object detection models with knowledge distillation. Neurips 2017.

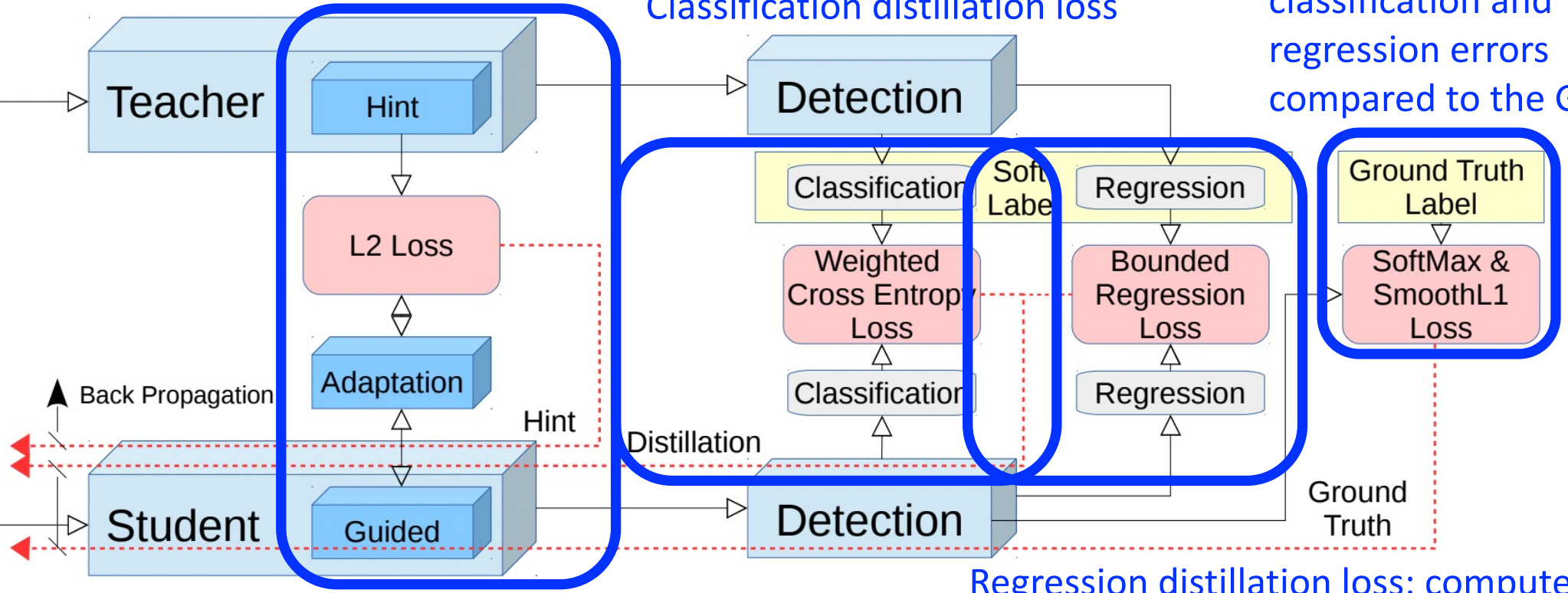
Approach for Creating Compact Student Model

A loss is computed to encourage the student's intermediate features to match those of the teacher

Classification distillation loss

Conventional loss computed for classification and regression errors compared to the GT





= backpropagation pathways

Regression distillation loss: computed if the student's distance to the GT exceeds the teacher's distance

Chen et al. Learning efficient object detection models with knowledge distillation. Neurips 2017.

Experiments

4 student models

3 teacher models

mAP scores for 5 datasets

Student	Model Info	Teacher	PASCAL	COCO@.5	COCO@[.5,.95]	KITTI	ILSVRC
Tucker	11M / 47ms	-	54.7	25.4	11.8	49.3	20.6
		AlexNet	57.6 (+2.9)	26.5 (+1.2)	12.3 (+0.5)	51.4 (+2.1)	23.6 (+1.3)
		VGGM	58.2 (+3.5)	26.4 (+1.1)	12.2 (+0.4)	51.4 (+2.1)	23.9 (+1.6)
		VGG16	59.4 (+4.7)	28.3 (+2.9)	12.6 (+0.8)	53.7 (+4.4)	24.4 (+2.1)
AlexNet	62M / 74ms	-	57.2	32.5	15.8	55.1	27.3
		VGGM	59.2 (+2.0)	33.4 (+0.9)	16.0 (+0.2)	56.3 (+1.2)	28.7 (+1.4)
		VGG16	60.1 (+2.9)	35.8 (+3.3)	16.9 (+1.1)	58.3 (+3.2)	30.1 (+2.8)
VGGM	80M / 86ms	-	59.8	33.6	16.1	56.7	31.1
		VGG16	63.7 (+3.9)	37.2 (+3.6)	17.3 (+1.2)	58.6 (+2.3)	34.0 (+2.9)
VGG16	138M / 283ms	-	70.4	45.1	24.2	59.2	35.6

- means no distillation or, in other words, trained from scratch

For all student-teacher pairs, knowledge distillation not only yields more compact and faster models but also leads to performance improvements

Chen et al. Learning efficient object detection models with knowledge distillation. Neurips 2017.

Experiments

4 student models

3 teacher models

mAP scores for 5 datasets

Student	Model Info	Teacher	PASCAL	COCO@.5	COCO@[.5,.95]	KITTI	ILSVRC
Tucker	11M / 47ms	-	54.7	25.4	11.8	49.3	20.6
		AlexNet	57.6 (+2.9)	26.5 (+1.2)	12.3 (+0.5)	51.4 (+2.1)	23.6 (+1.3)
		VGGM	58.2 (+3.5)	26.4 (+1.1)	12.2 (+0.4)	51.4 (+2.1)	23.9 (+1.6)
		VGG16	59.4 (+4.7)	28.3 (+2.9)	12.6 (+0.8)	53.7 (+4.4)	24.4 (+2.1)
AlexNet	62M / 74ms	-	57.2	32.5	15.8	55.1	27.3
		VGGM	59.2 (+2.0)	33.4 (+0.9)	16.0 (+0.2)	56.3 (+1.2)	28.7 (+1.4)
		VGG16	60.1 (+2.9)	35.8 (+3.3)	16.9 (+1.1)	58.3 (+3.2)	30.1 (+2.8)
VGGM	80M / 86ms	-	59.8	33.6	16.1	56.7	31.1
		VGG16	63.7 (+3.9)	37.2 (+3.6)	17.3 (+1.2)	58.6 (+2.3)	34.0 (+2.9)
VGG16	138M / 283ms	-	70.4	45.1	24.2	59.2	35.6

- means no distillation or, in other words, trained from scratch

Why do you think there are performance improvements from model compression?

Chen et al. Learning efficient object detection models with knowledge distillation. Neurips 2017.

Knowledge Distillation for Vision

- Object detection

- Image classification with attention transfer

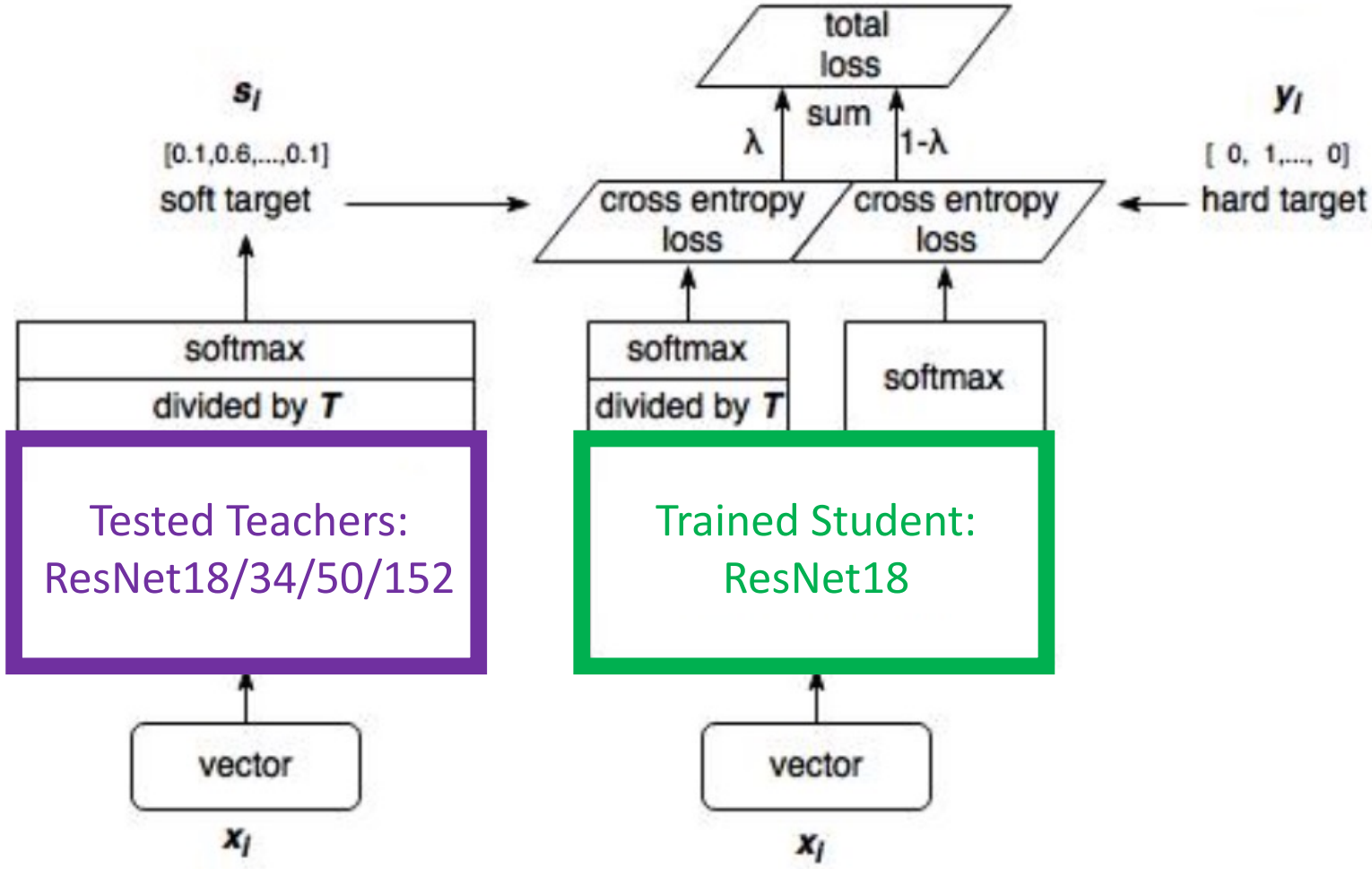
Recall Task: Predict Category from 1000 Options

- Evaluation metric: % correct (top-1 and top-5 predictions)
- Dataset: ~1.5 million images
- Source: images scraped from search engines, such as Flickr, and labeled by crowdworkers



J. Deng, W. Dong, R. Socher, L. Li, K. Li and L. Fei-Fei. ImageNet: A Large-Scale Hierarchical Image Database. 2009

Experiment: Do Bigger, More Accurate Models Make Better Teachers?



Cho and Hariharan. On the Efficacy of Knowledge Distillation. ICCV 2019

Figure source: https://blog.csdn.net/qq_22749699/article/details/79460817

Experiment: Do Bigger, More Accurate Models Make Better Teachers?

(% = Top-1 error rates)

Teacher	Teacher Error (%)	Student Error (%)
ResNet18	30.24	30.57
ResNet34	26.70	30.79
ResNet50	23.85	30.95

What is the student's performance trend from larger, more accurate teachers?

Cho and Hariharan. On the Efficacy of Knowledge Distillation. ICCV 2019

Experiment: Do Bigger, More Accurate Models Make Better Teachers?

(% = Top-1 error rates)

Teacher	Teacher Error (%)	Student Error (%)
-	-	30.24
ResNet18	30.24	30.57
ResNet34	26.70	30.79
ResNet50	23.85	30.95

Student performance not only drops for larger teachers but the models distilled from teachers perform worse than training the student from scratch!

Cho and Hariharan. On the Efficacy of Knowledge Distillation. ICCV 2019

Experiment: Why Might Student Performance Drop as Teacher Size Grows?

1. More accurate models are more confident and so need higher temperatures to learn the “dark knowledge” of category relationships

2. Student mimics teacher but the loss function is mismatched from the evaluation metric

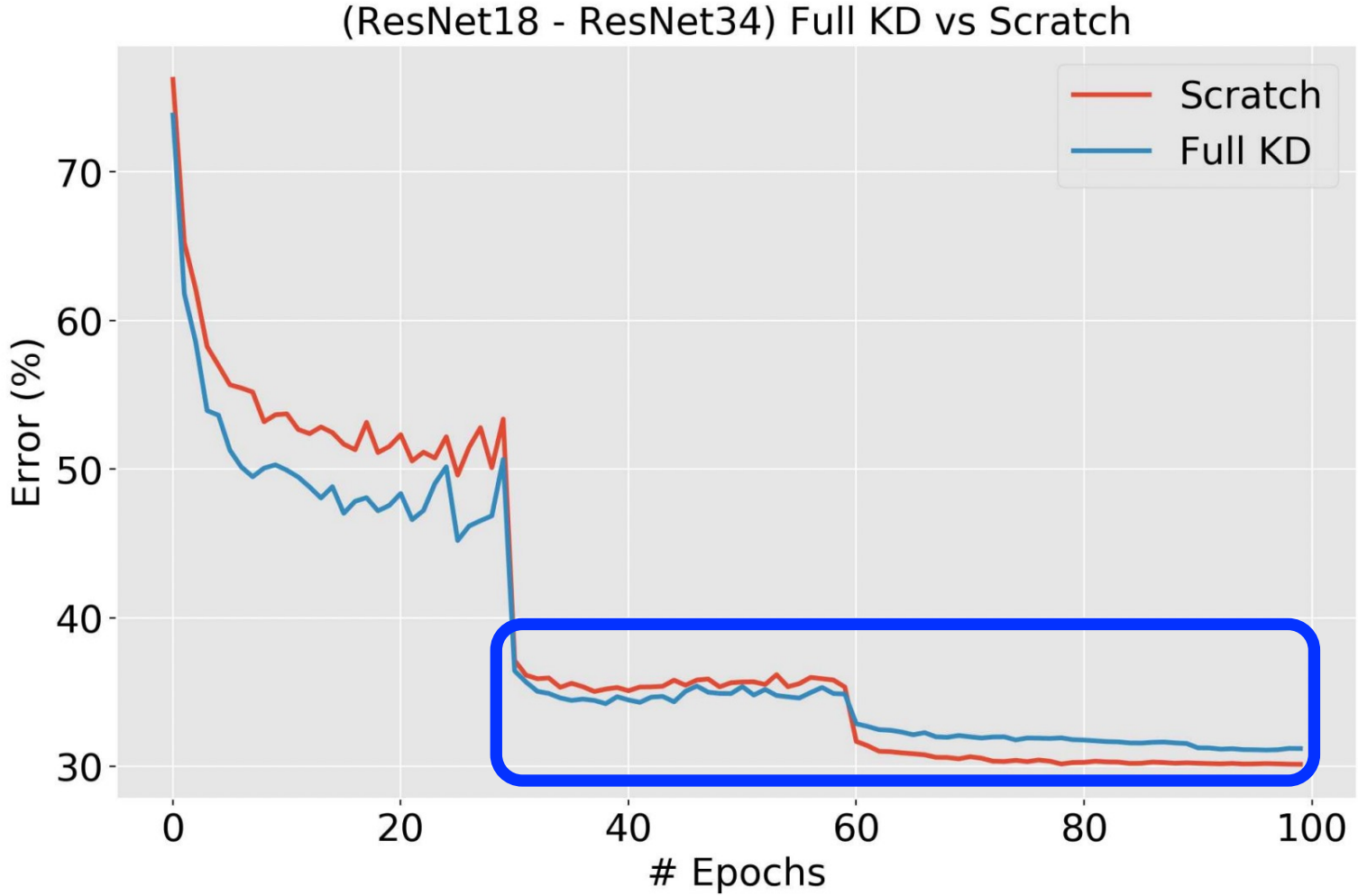
3. Student fails to accurately mimic teacher

Experimental analysis suggests this is the reason

Cho and Hariharan. On the Efficacy of Knowledge Distillation. ICCV 2019

Experiment: Why Might Student Students Fail to Mimic Teachers?

Hypothesis: student is underfitting because of lower capacity and so “minimizing one loss (KD loss) at the expense of the other (cross entropy loss)”

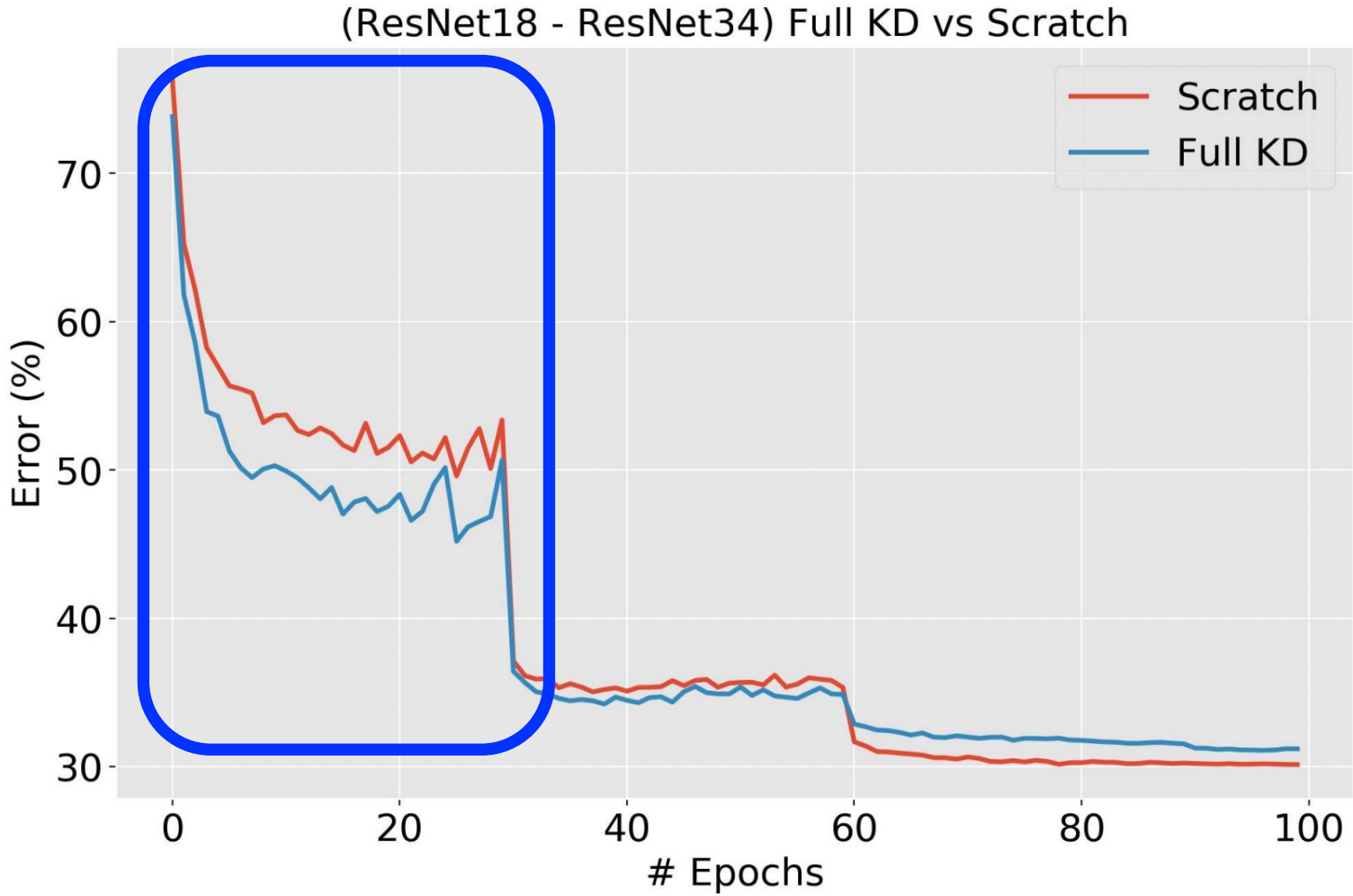


Cho and Hariharan. On the Efficacy of Knowledge Distillation. ICCV 2019

Experiment: Why Might Student Students Fail to Mimic Teachers?

How to overcome this issue?

- Early stopping with KD loss (ESKD) to leverage its benefit at the start of training



Cho and Hariharan. On the Efficacy of Knowledge Distillation. ICCV 2019

Experiments: How Does ESKD Compare To Training A Student from Scratch?

Teacher	Top-1 Error (%, Test)
ResNet18	30.57
ResNet18 (ES KD)	29.01
ResNet34	30.79
ResNet34 (ES KD)	29.16
ResNet50	30.95
ResNet50 (ES KD)	29.35

Training a model with early stopping knowledge distillation loss leads to better results than training from scratch!

Cho and Hariharan. On the Efficacy of Knowledge Distillation. ICCV 2019

Experiments: Does ESKD with Bigger, More Accurate Models Make Better Teachers?

Teacher	Top-1 Error (%, Test)
ResNet18	30.57
ResNet18 (ES KD)	29.01
ResNet34	30.79
ResNet34 (ES KD)	29.16
ResNet50	30.95
ResNet50 (ES KD)	29.35

No; the student may still be struggling with underfitting due to an insufficient representational capacity

Cho and Hariharan. On the Efficacy of Knowledge Distillation. ICCV 2019

Experiments: To Address The Capacity Problem Why Not Instead Distill to Intermediate Sizes?

Performs almost identically to a model that is distilled directly from a large to small size; does not address the core problem:

The student must be in the solution space of the teacher

Cho and Hariharan. On the Efficacy of Knowledge Distillation. ICCV 2019

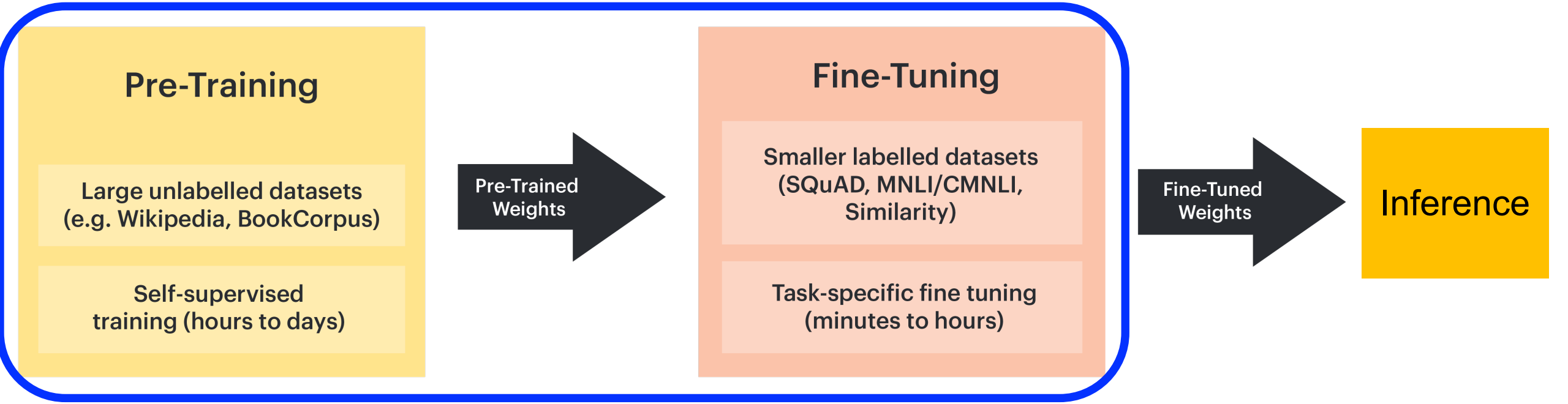
Knowledge Distillation for Vision

- Object detection
- Image classification with attention transfer

Today's Topics

- Motivation
- Key idea: knowledge distillation
- Knowledge distillation for CNNs (vision problems)
- Knowledge distillation for Transformers (language problems)
- Latex tutorial

Recall Popular Transformer Approach (e.g., BERT)

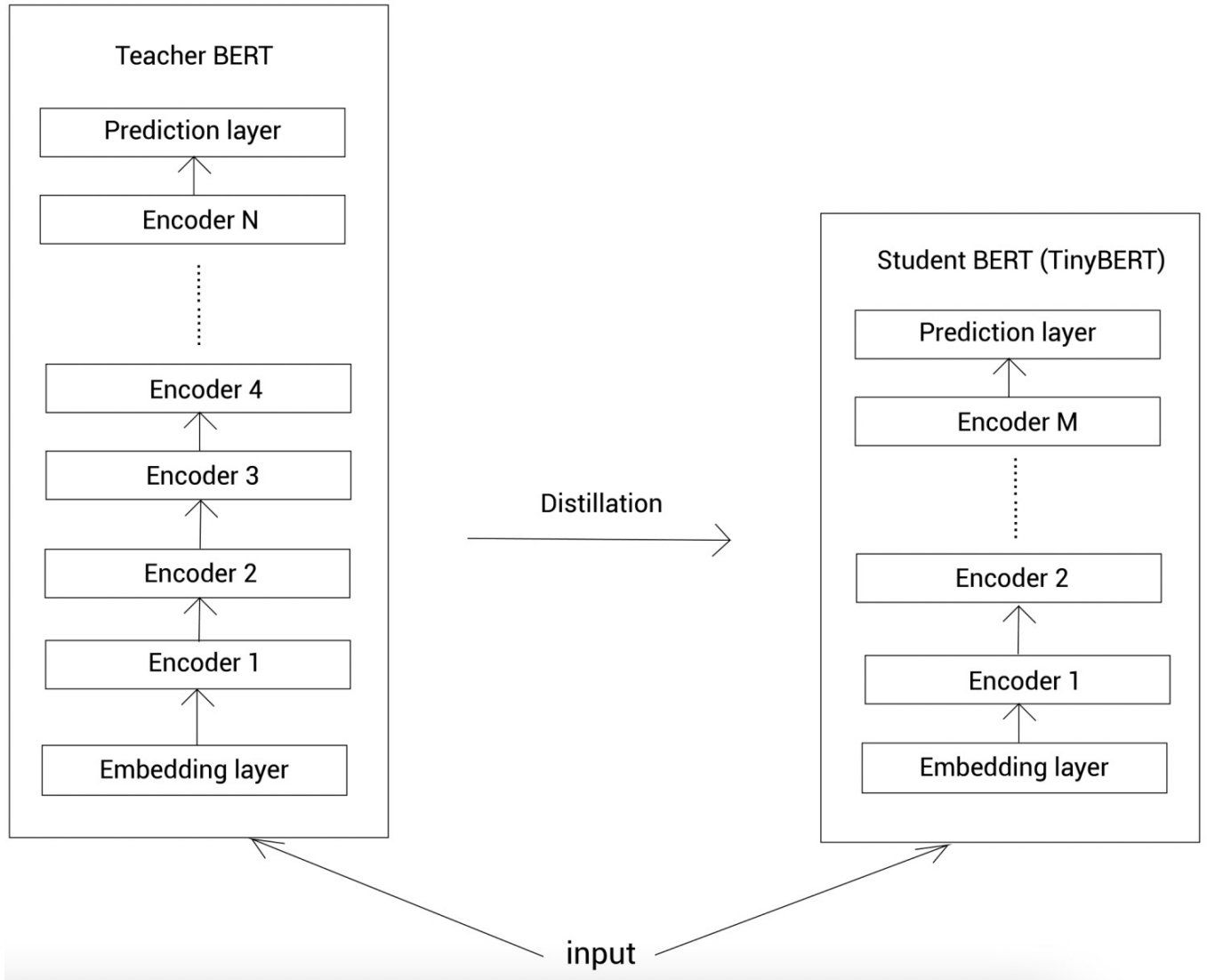


Knowledge distillation needed at both the pretraining and fine-tuning stages

https://docs.graphcore.ai/projects/bert-training/en/latest/bert.html

TinyBERT: Pre-training Knowledge Distillation

BERT-Base with 12 encoder layers and 12 attention heads that produces a 768 representation:

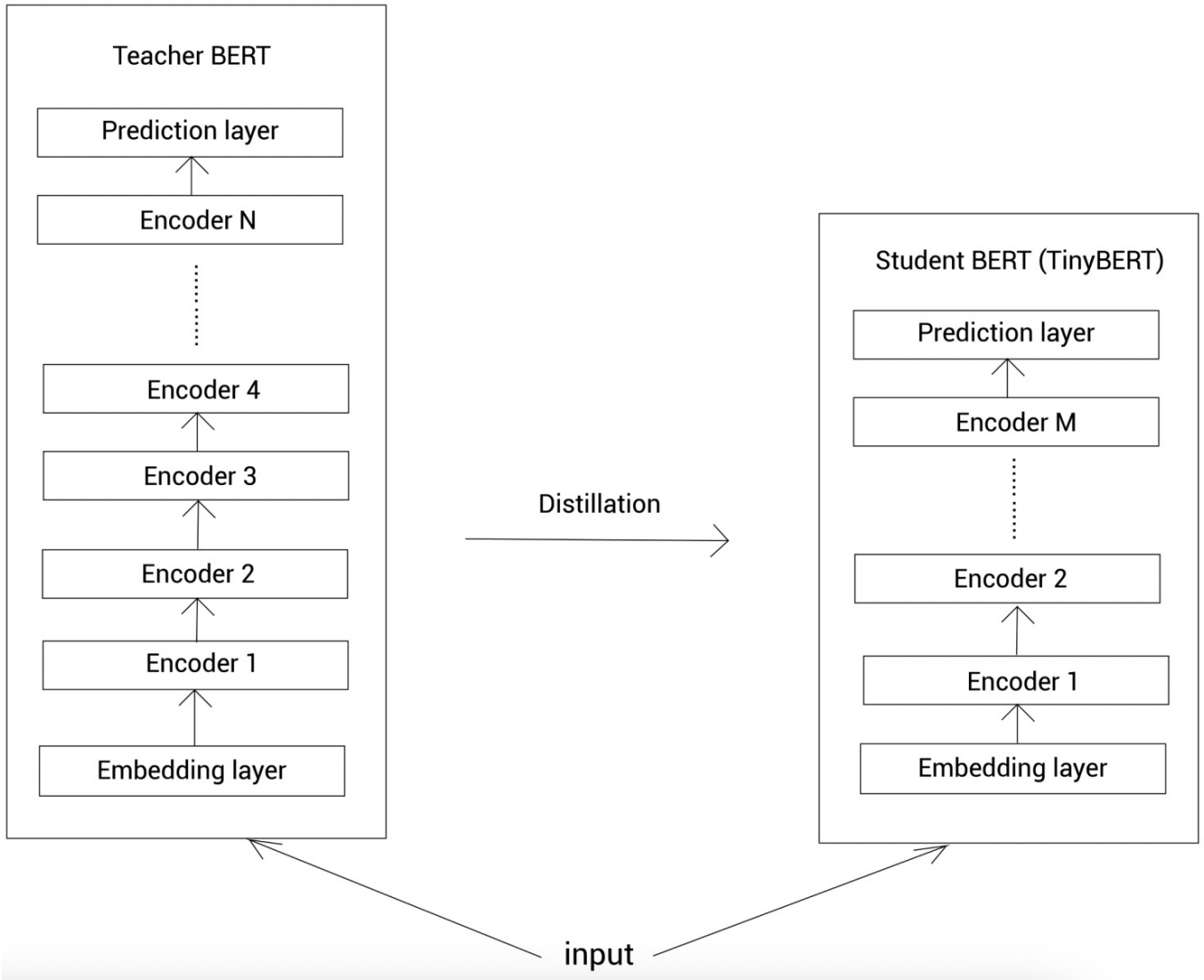


Same architecture but with 4 encoder layers and it produces a 312 representation

https://static.packt-cdn.com/downloads/9781838821593_ColorImages.pdf

TinyBERT: Pre-training Knowledge Distillation

109 million parameters

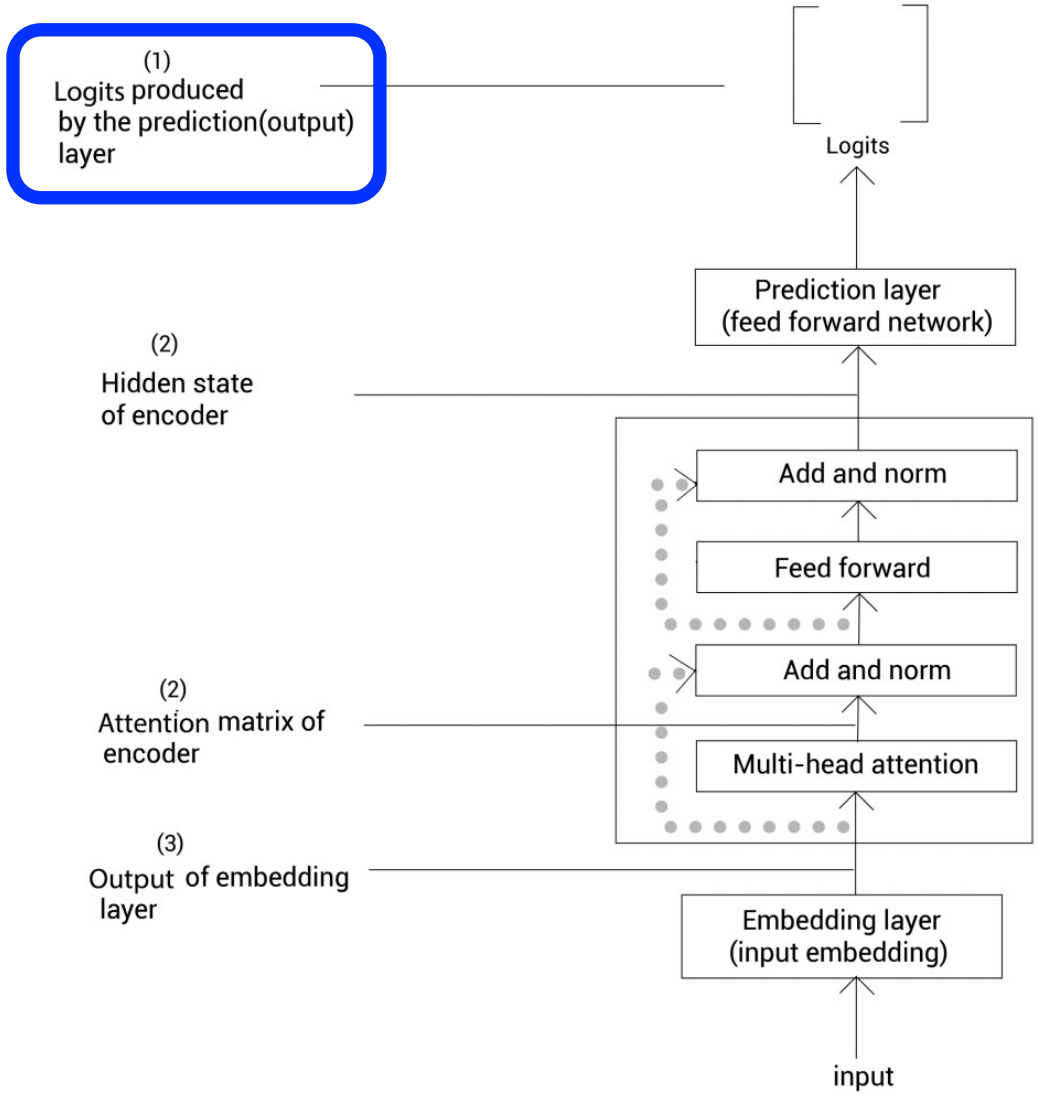


14.5 million parameters

https://static.packt-cdn.com/downloads/9781838821593_ColorImages.pdf

TinyBERT: Pre-training Knowledge Distillation

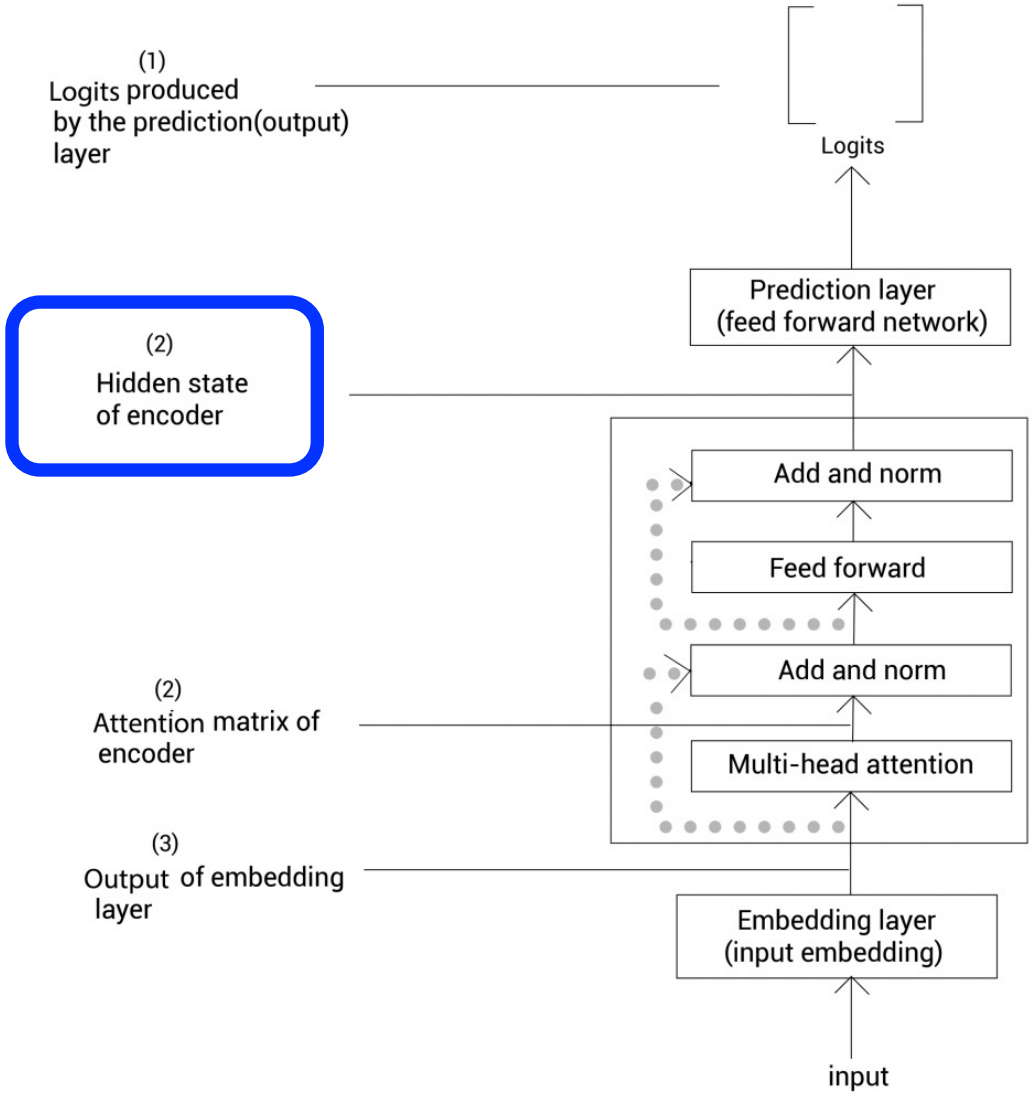
Student learns from teacher's output (i.e.., logits) and intermediate layers:



https://static.packt-cdn.com/downloads/9781838821593_ColorImages.pdf

TinyBERT: Pre-training Knowledge Distillation

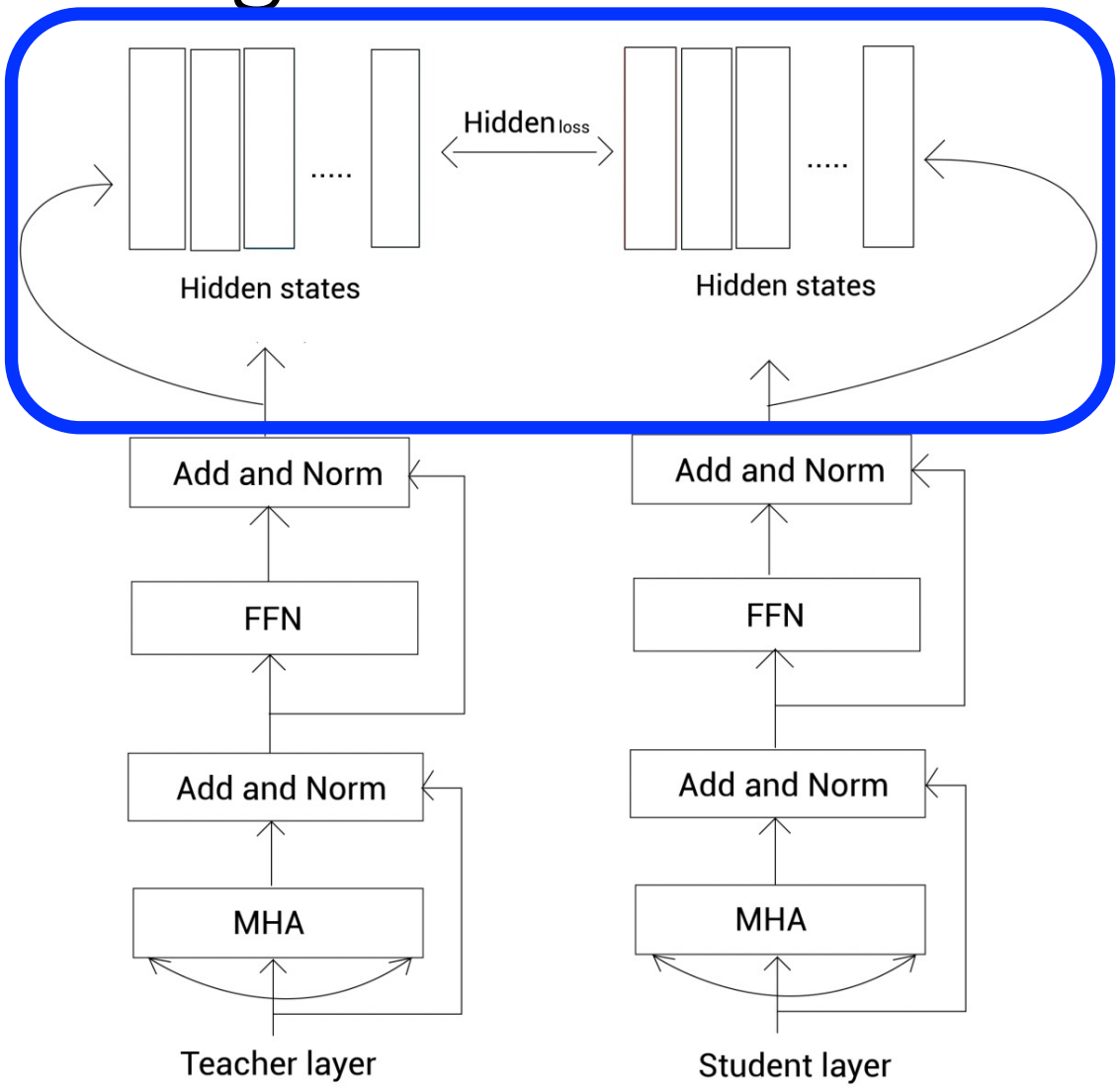
Student learns from teacher's output (i.e.., logits) and intermediate layers:



https://static.packt-cdn.com/downloads/9781838821593_ColorImages.pdf

TinyBERT: Pre-training Knowledge Distillation

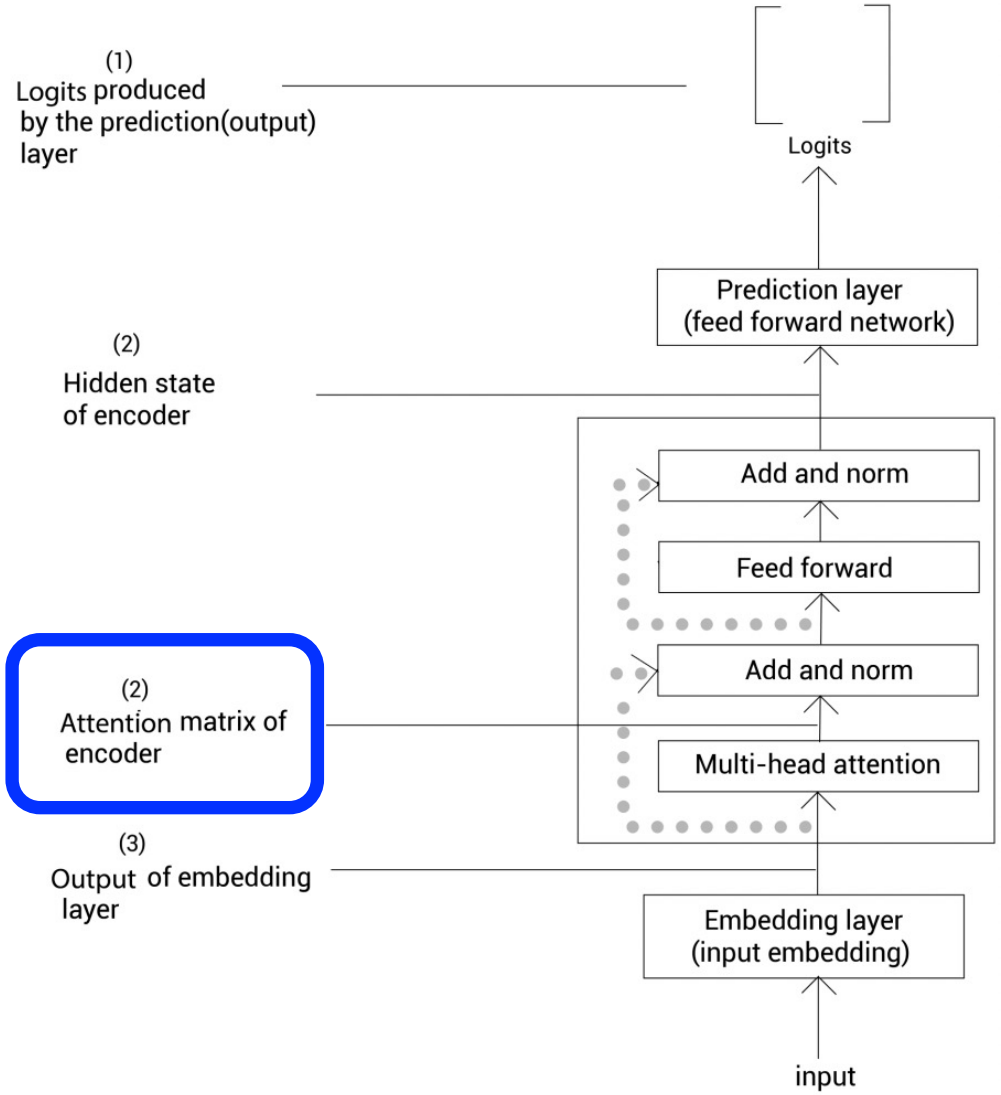
Mean squared error computed between encoded representations with weight matrix converting student's 312 representation to match teacher's 768 representation



https://static.packt-cdn.com/downloads/9781838821593_ColorImages.pdf

TinyBERT: Pre-training Knowledge Distillation

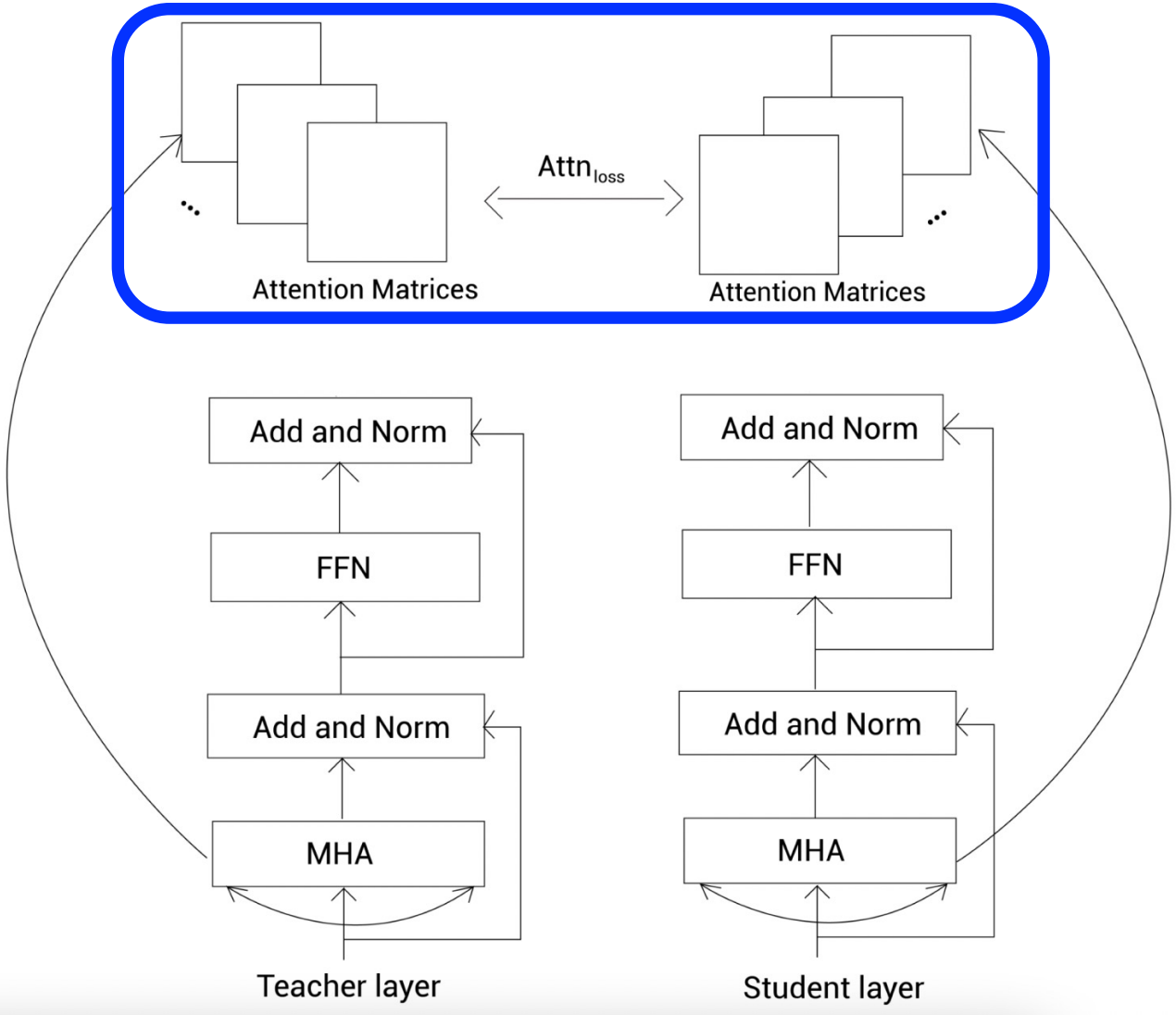
Student learns from teacher's output (i.e.., logits) and intermediate layers:



https://static.packt-cdn.com/downloads/9781838821593_ColorImages.pdf

TinyBERT: Pre-training Knowledge Distillation

Mean squared error computed between all attention matrices of the teacher and student

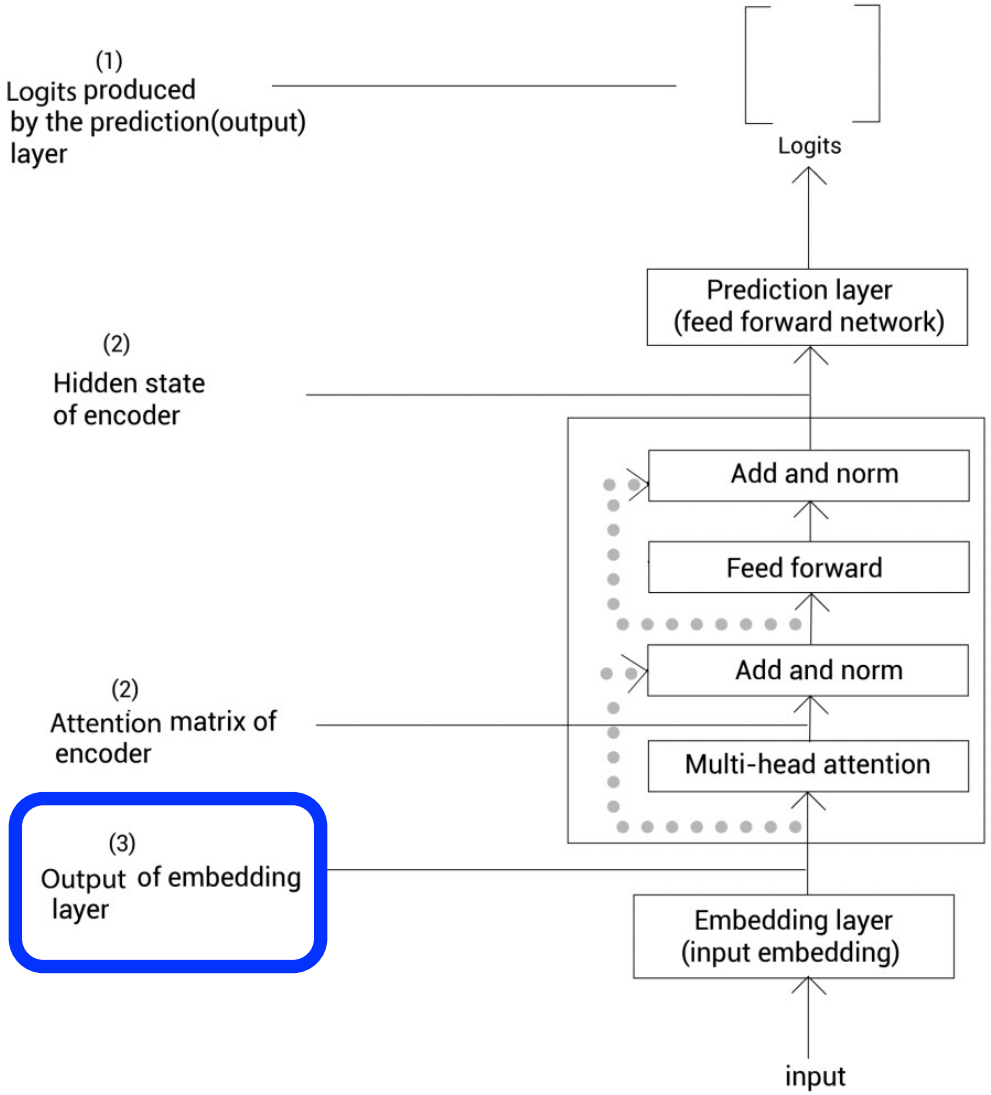


https://static.packt-cdn.com/downloads/9781838821593_ColorImages.pdf

TinyBERT: Pre-training Knowledge Distillation

Student learns from teacher's output (i.e.., logits) and intermediate layers:

Mean squared error computed between embedding representations with weight matrix converting student's representation to match teacher's representation

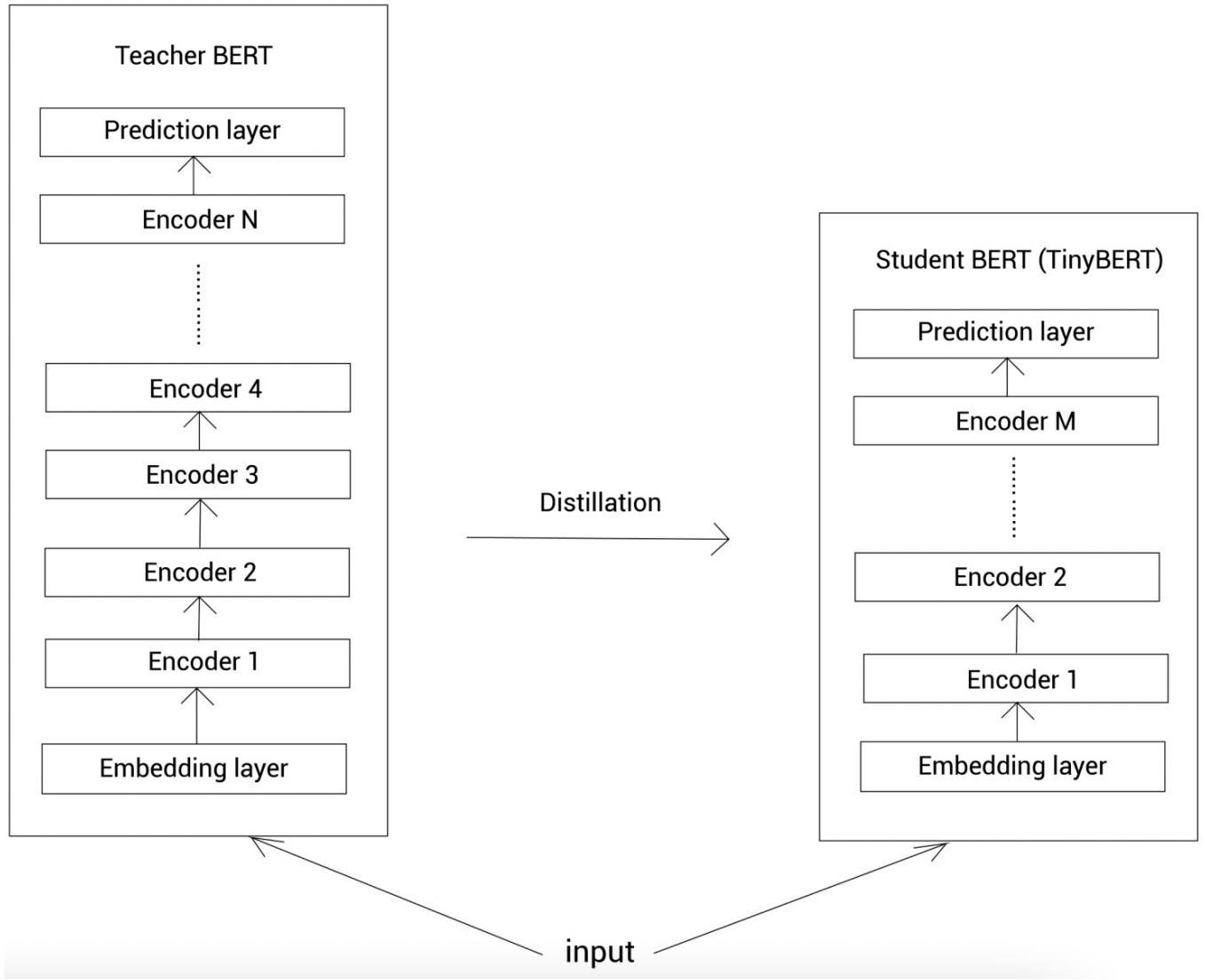


https://static.packt-cdn.com/downloads/9781838821593_ColorImages.pdf

TinyBERT: Pre-training Knowledge Distillation

Training data matches that used for BERT: Wikipedia and Toronto BookCorpus

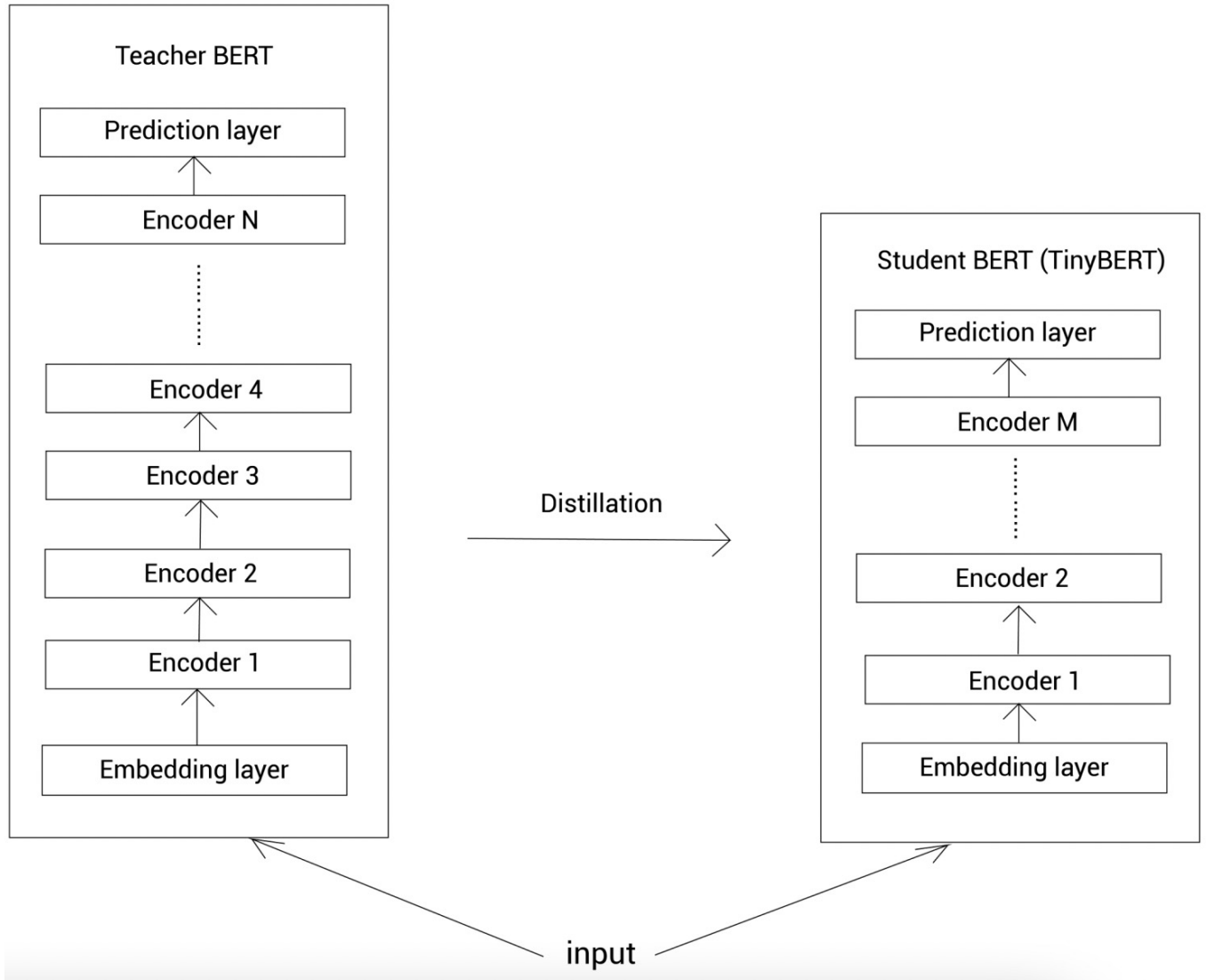
Training run for three epochs with errors of intermediate layers



https://static.packt-cdn.com/downloads/9781838821593_ColorImages.pdf

TinyBERT: Fine-tuning Knowledge Distillation

Fine-tuned BERT
for the target task

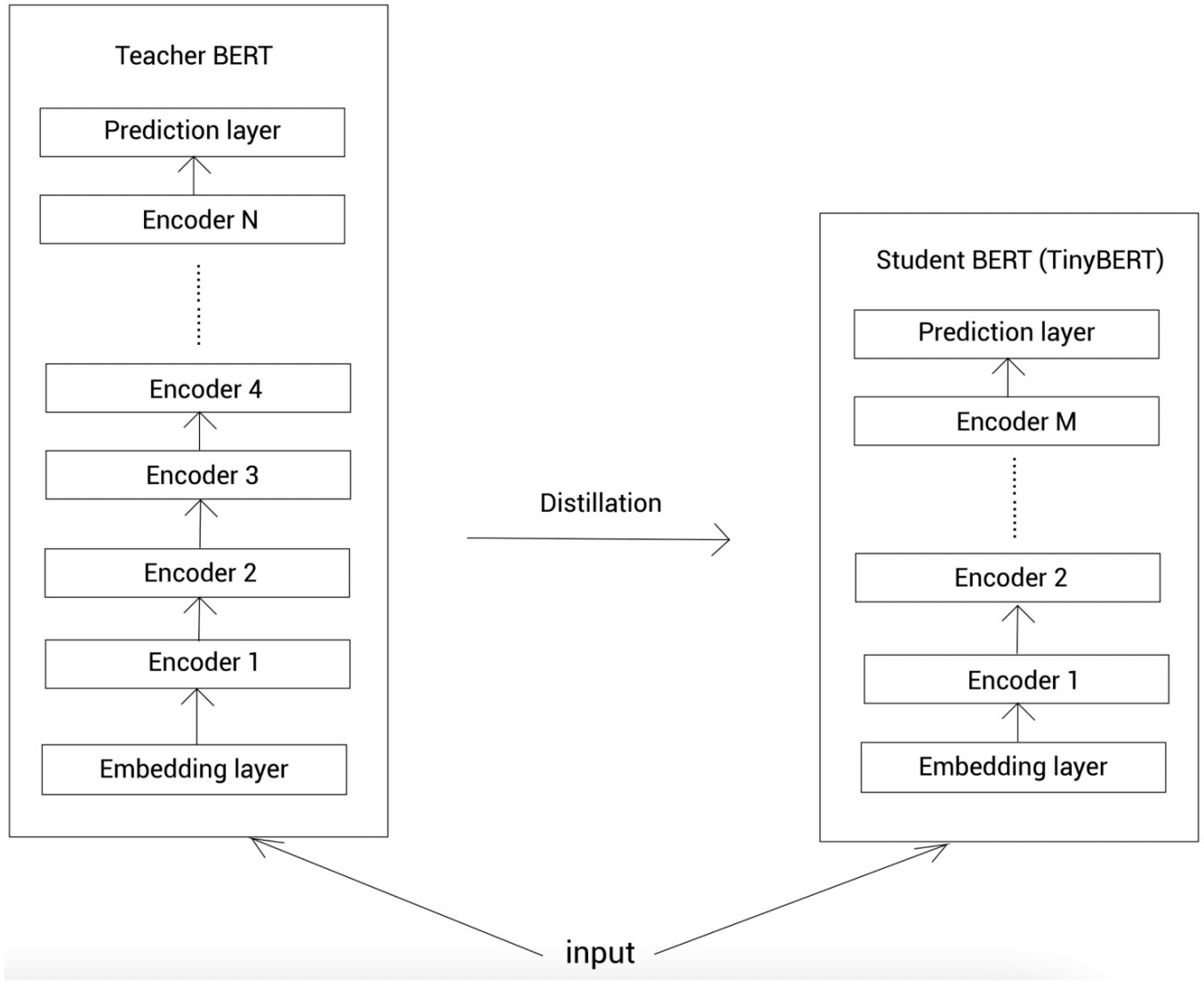


TinyBERT distilled
from pretrained BERT

https://static.packt-cdn.com/downloads/9781838821593_ColorImages.pdf

TinyBERT: Fine-tuning Knowledge Distillation

Training run for 20 epochs with errors of intermediate layers and then for 30 epochs using errors from the prediction layer



https://static.packt-cdn.com/downloads/9781838821593_ColorImages.pdf

TinyBERT: Fine-tuning Knowledge Distillation

Compared to BERT-base, TinyBERT achieves nearly comparable performance while containing ~13.3% of the parameters and needing only ~10.6% of inference time

https://static.packt-cdn.com/downloads/9781838821593_ColorImages.pdf

Today's Topics

- Motivation
- Key idea: knowledge distillation
- Knowledge distillation for CNNs (vision problems)
- Knowledge distillation for Transformers (language problems)
- Latex tutorial

Today’s Topics

- Motivation
- Key idea: knowledge distillation
- Knowledge distillation for CNNs (vision problems)
- Knowledge distillation for Transformers (language problems)
- Latex tutorial


The End